



Módulo 2 – Machine Learning

Bloque Temático 3 y 4

- Repaso:
 - Gradiente Descendiente
 - Backpropagation.
- Práctica.

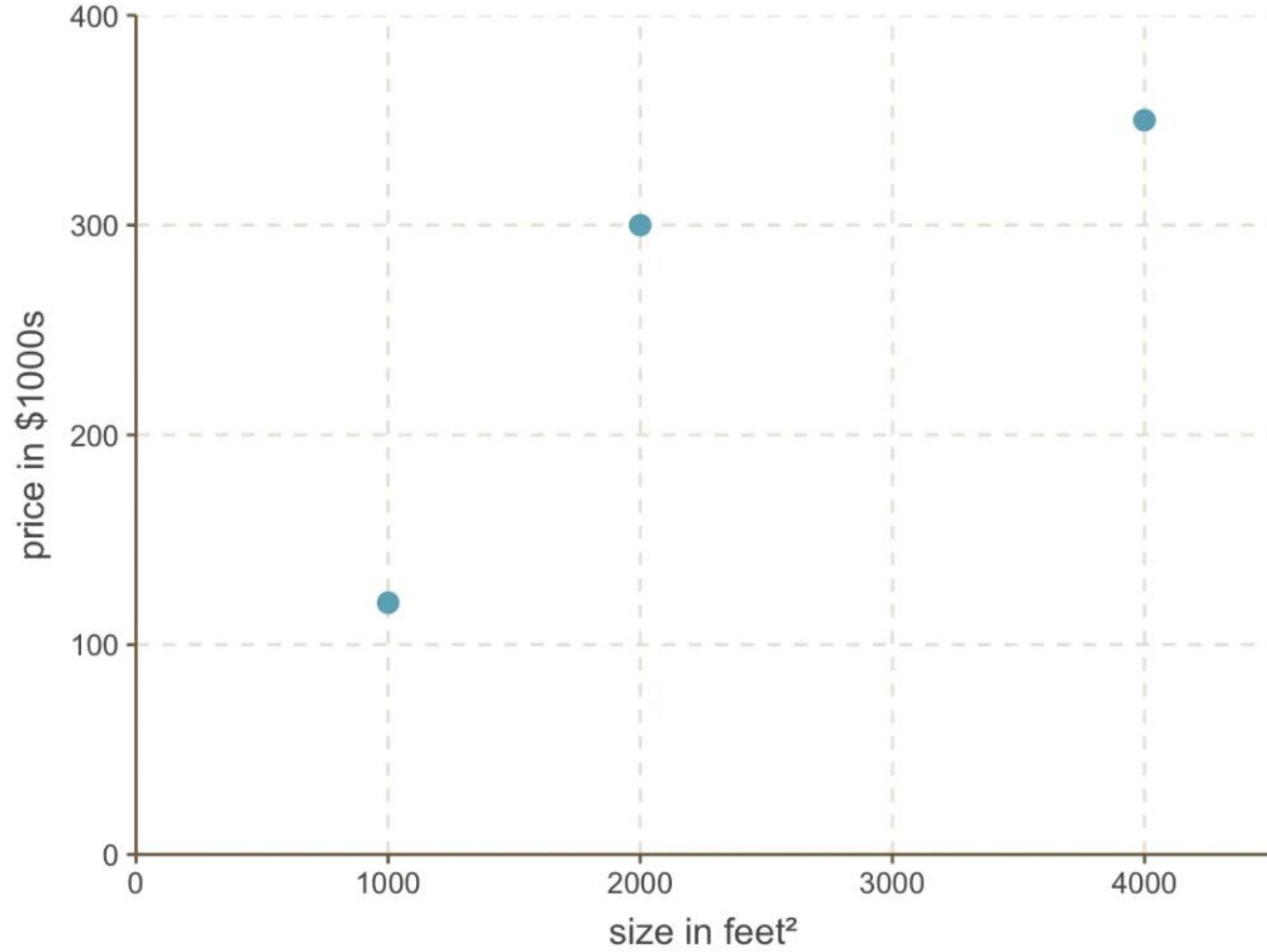


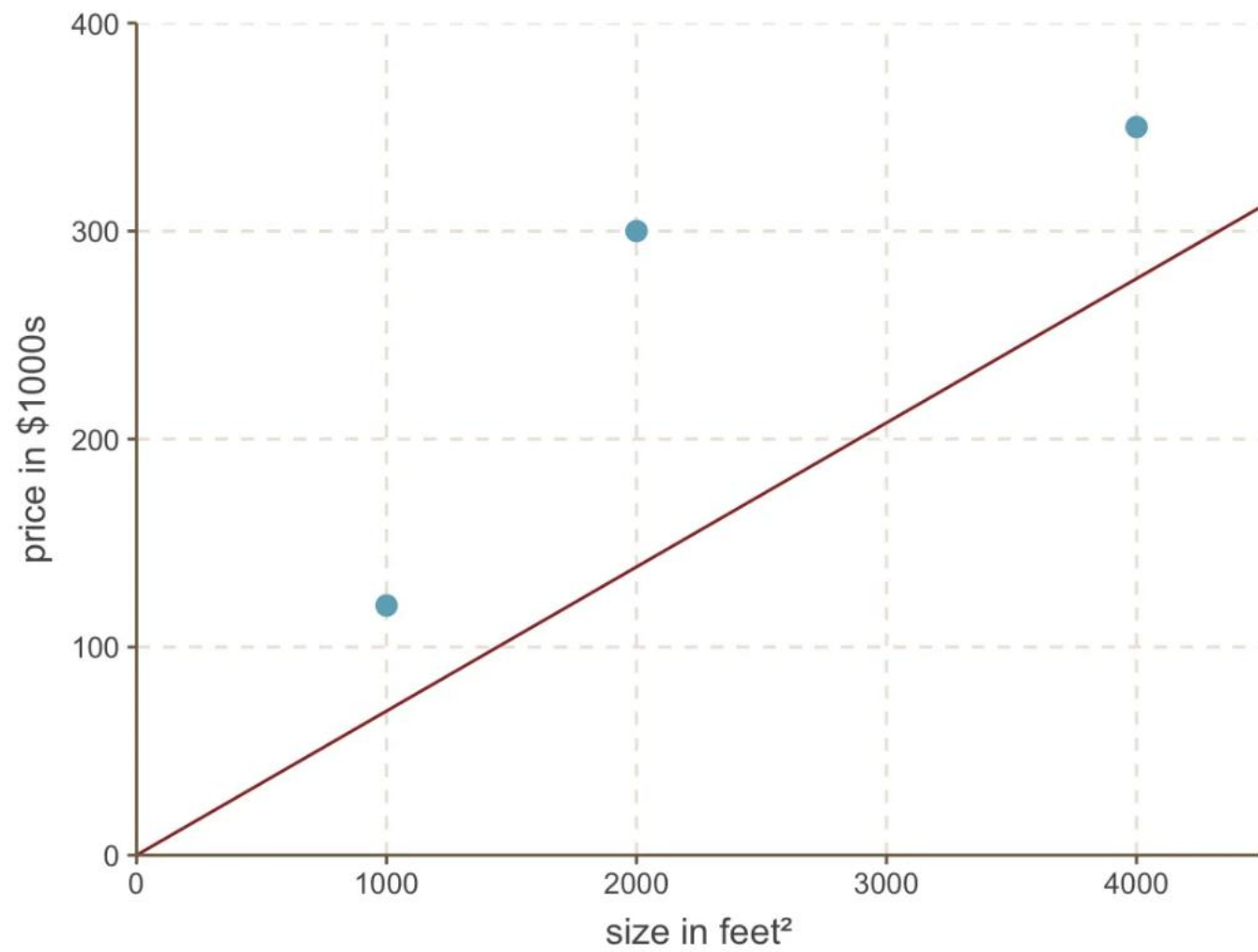
Introducción a Redes Neuronales

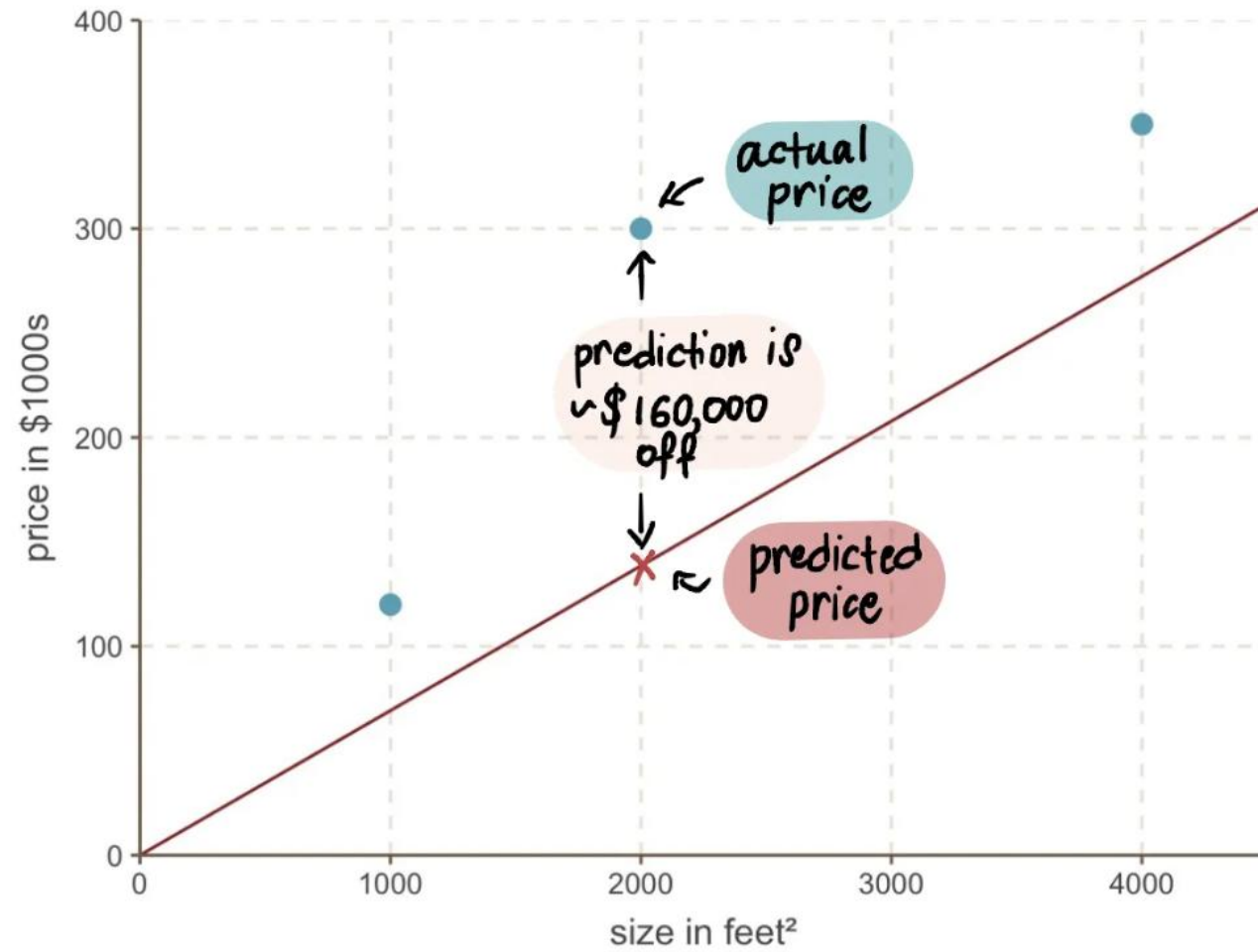
Gradiente Descendiente

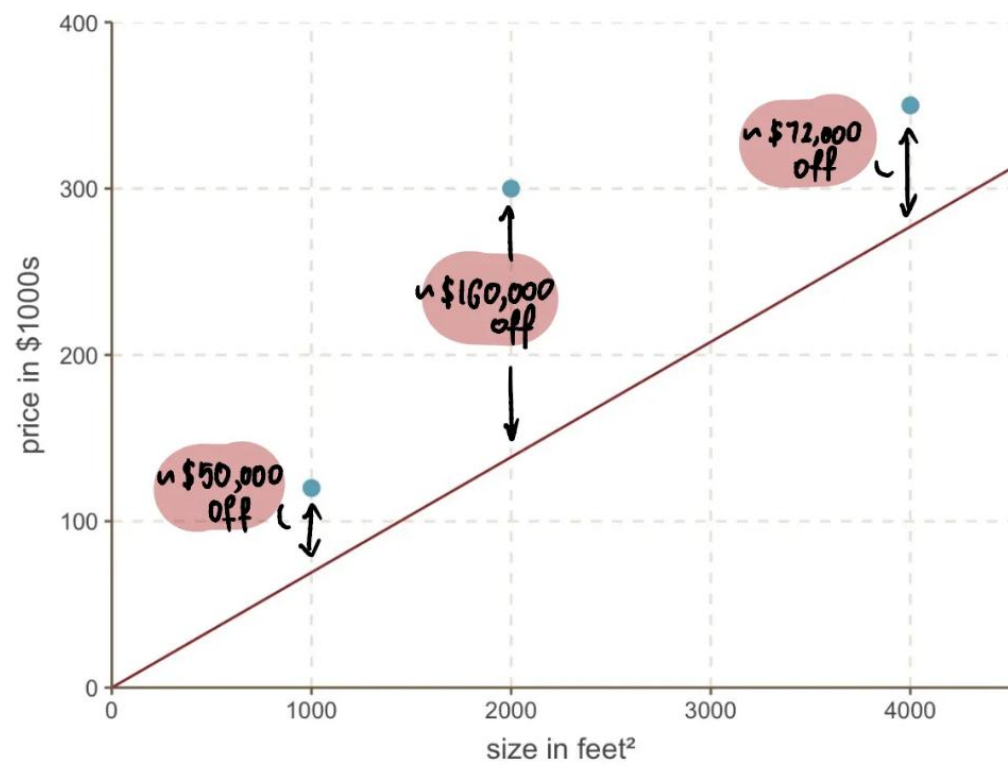


<u>house size</u>	<u>price in \$1000s</u>
1000 feet ²	120
2000 feet ²	300
3000 feet ²	350



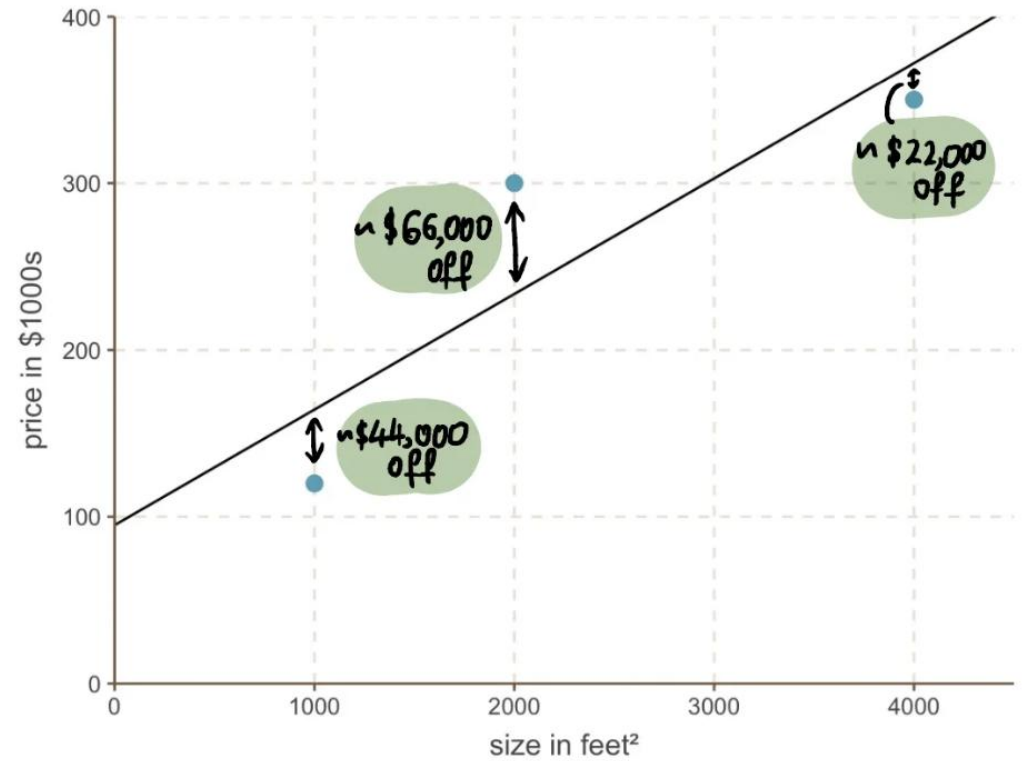
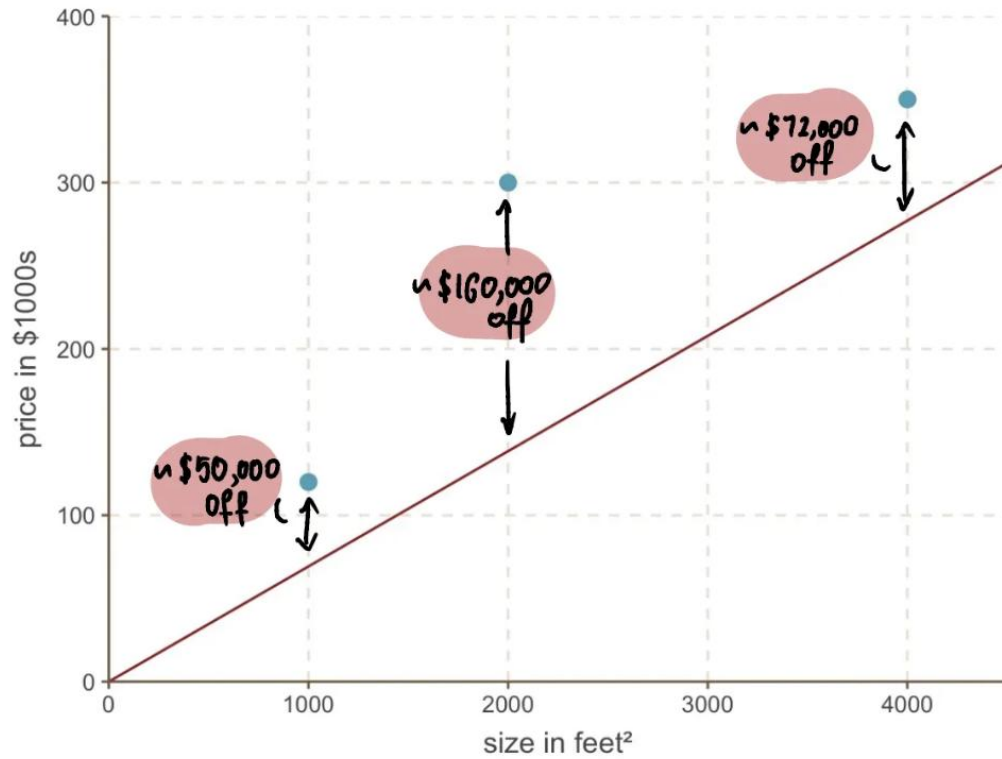






$$y = wx + b$$

$\swarrow$ = 0
 $\searrow$ $\neq 0$



② the absolute difference between the actual and predicted house price

① predicted house price

$$MAE = \frac{1}{n} \sum_{i=1}^n |y - \hat{y}_i|$$

④ calculate the average

③ sum of all the absolute differences

② the absolute difference between the actual and predicted house price

① predicted house price

$$MAE = \frac{1}{n} \sum_{i=1}^n |y - \hat{y}_i|$$

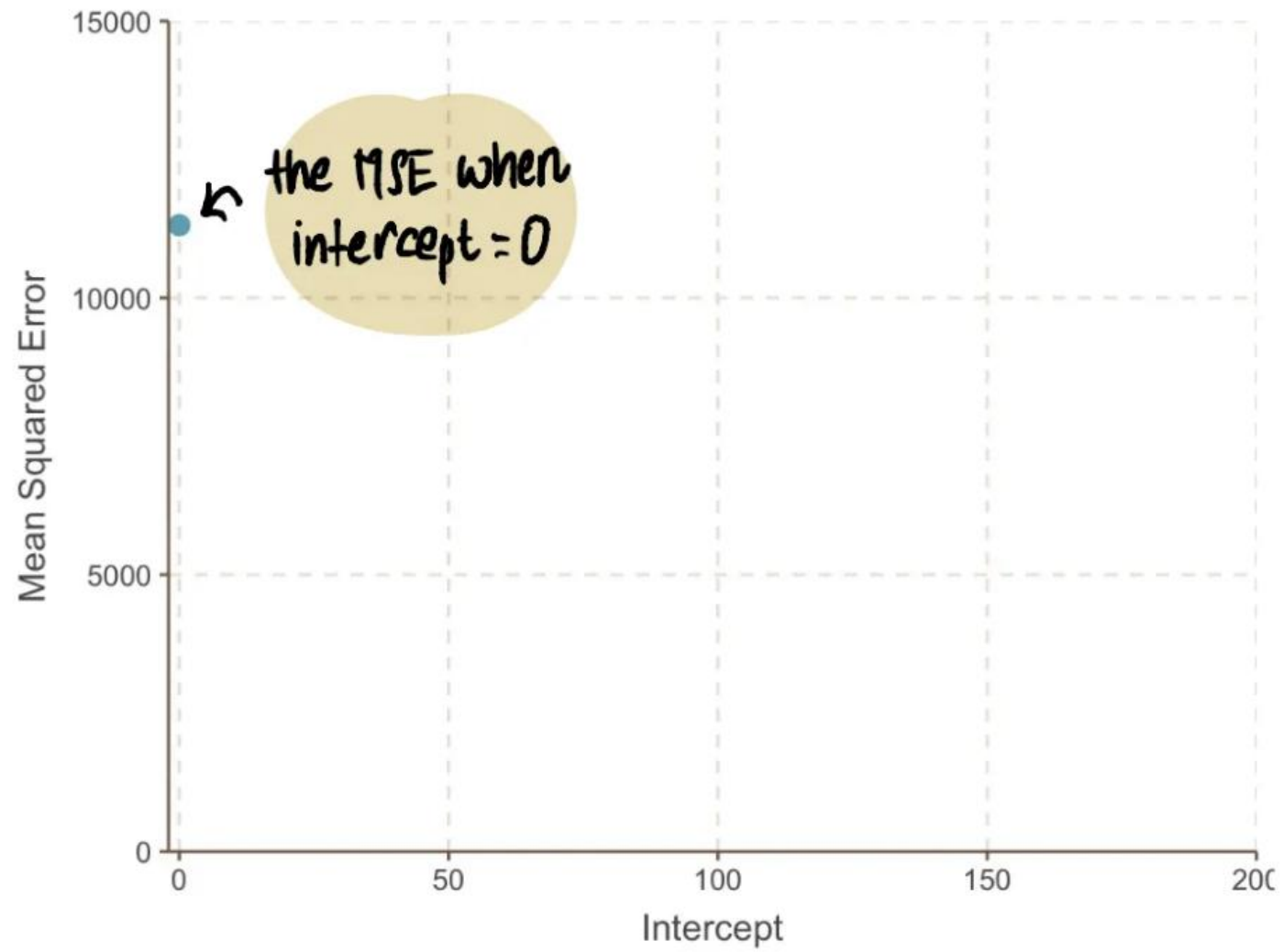
④ calculate the average

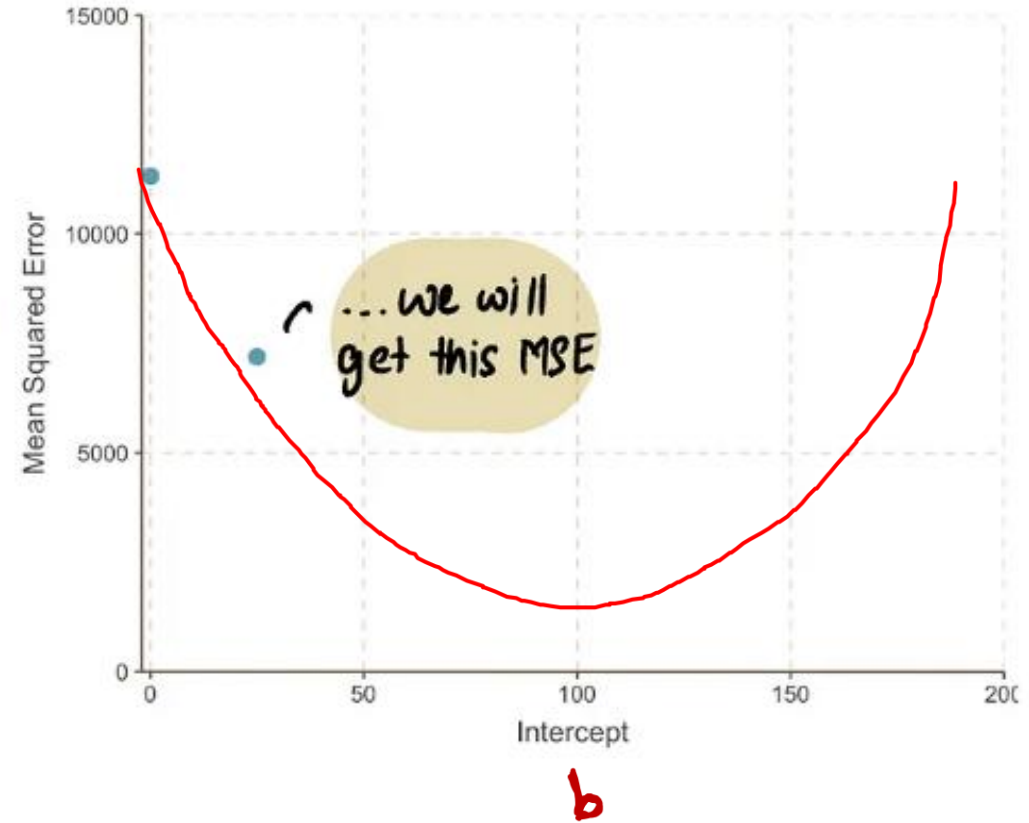
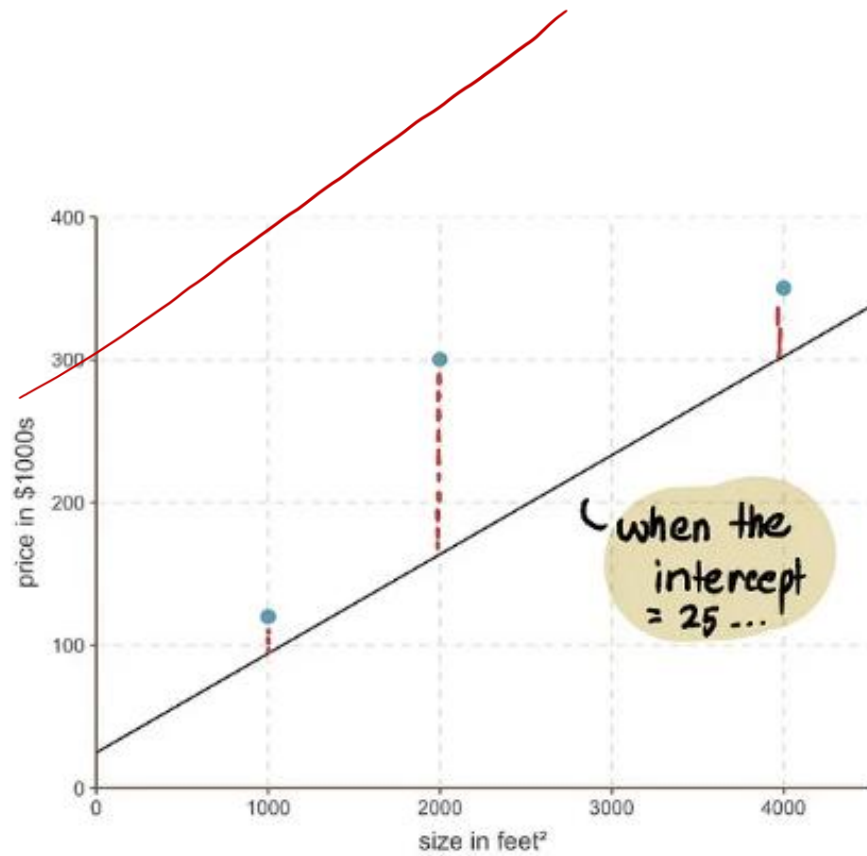
③ sum of all the absolute differences

actual house price

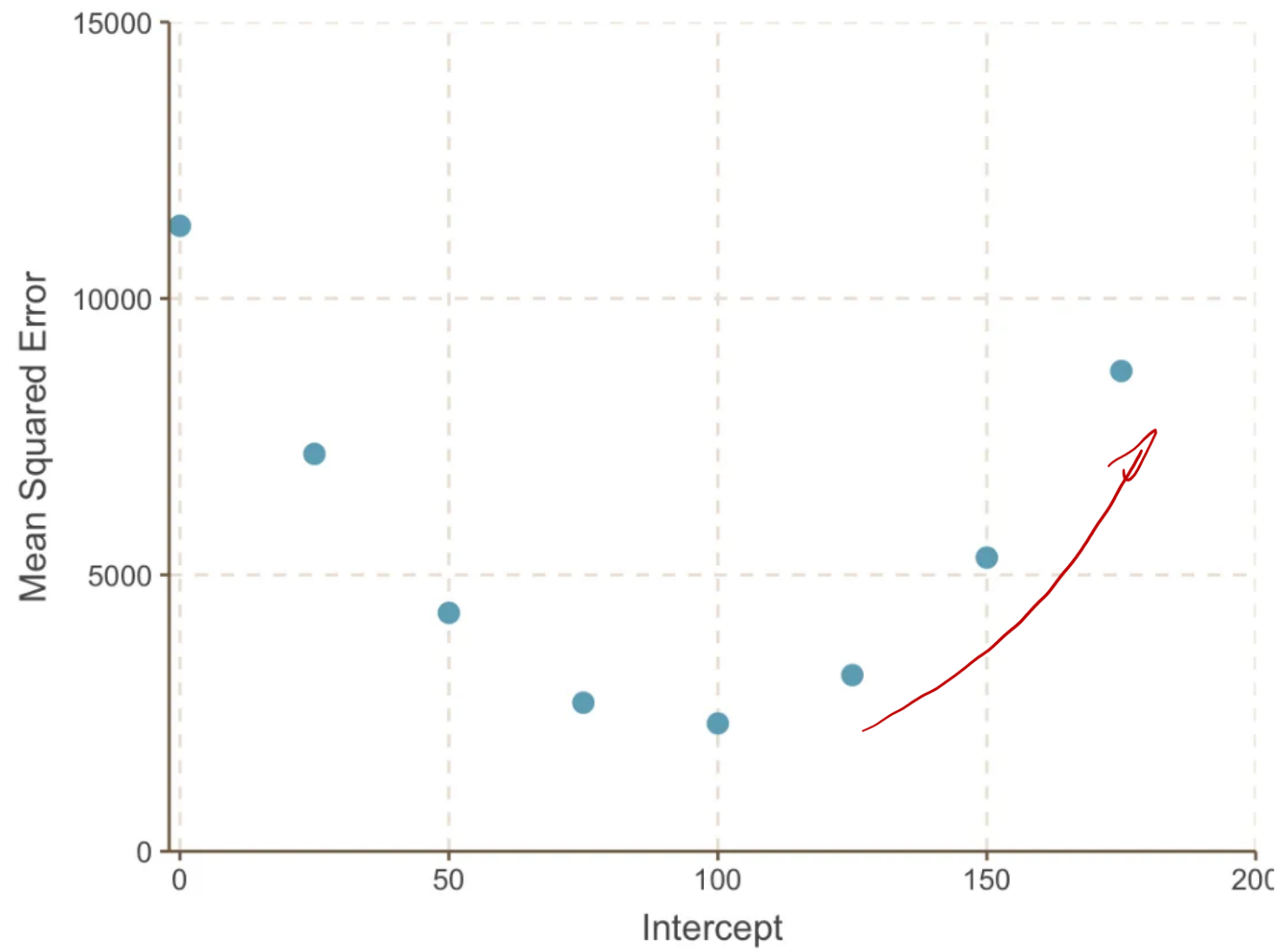
predicted house price when Intercept = 0

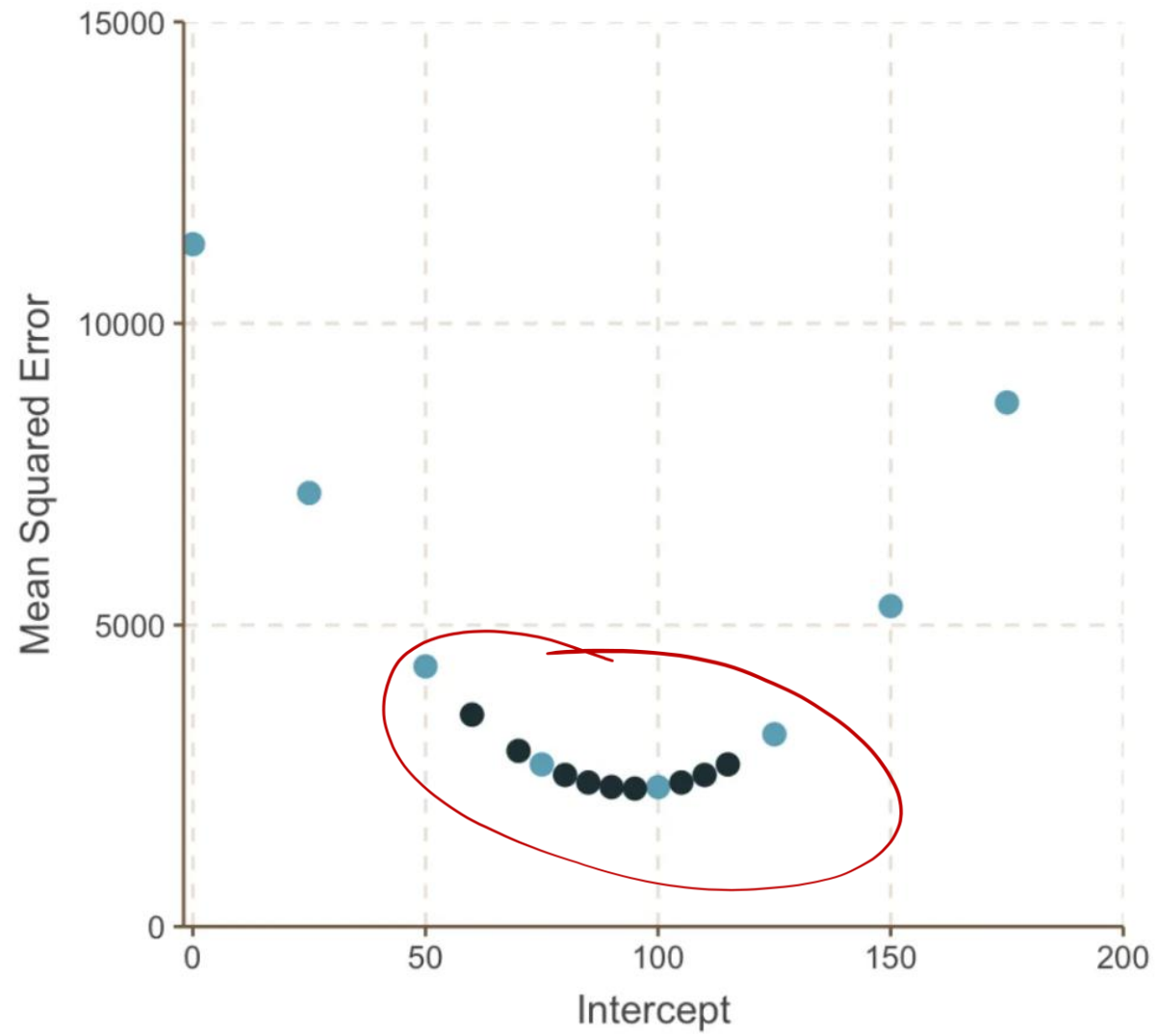
$$\begin{aligned} MSE &= \frac{1}{3} [(120 - (0 + 0.069 \times 1000))^2 + (300 - (0 + 0.069 \times 2000))^2 \\ &\quad + (350 - (0 + 0.069 \times 4000))^2] \\ &= 11313 \end{aligned}$$



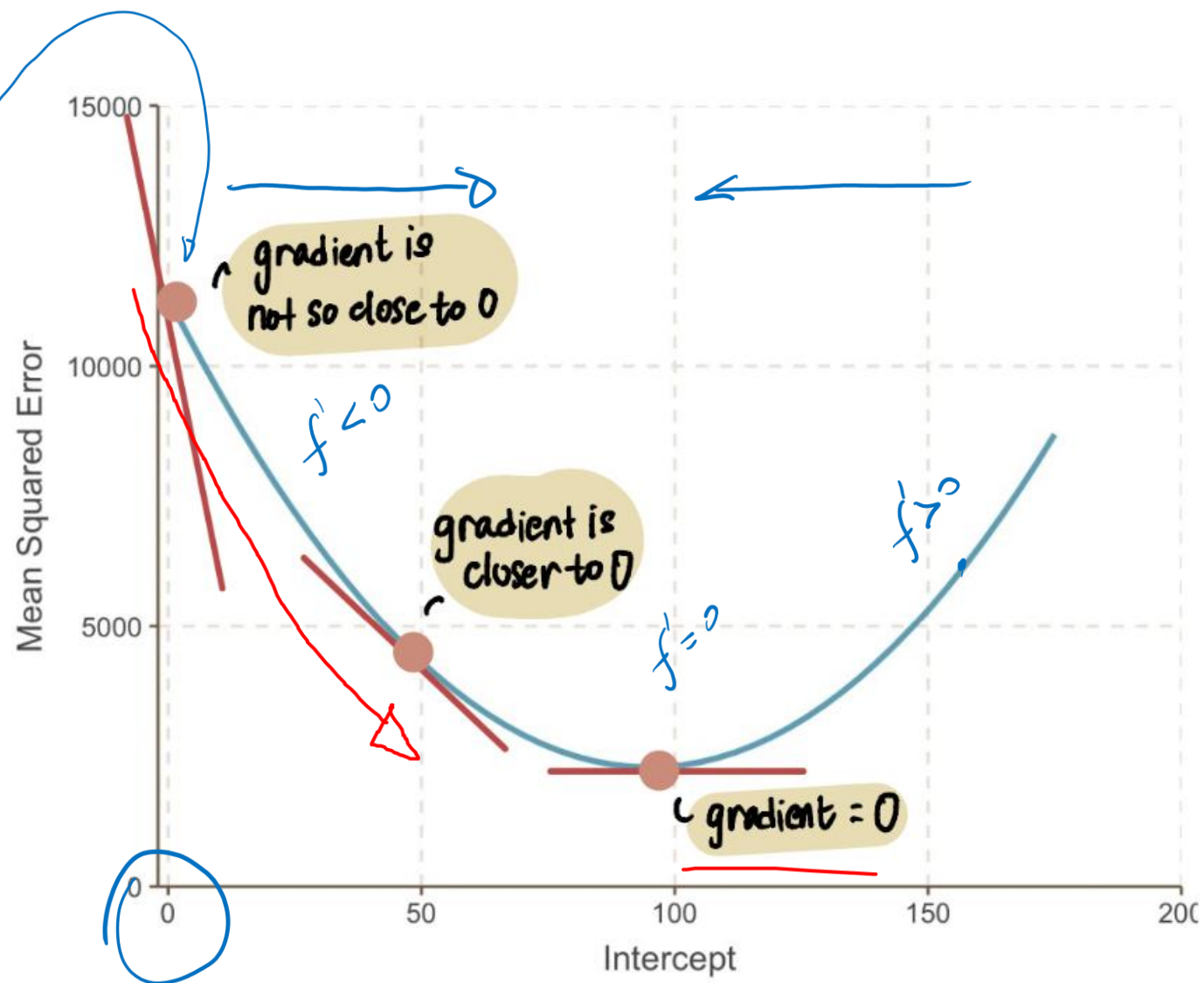
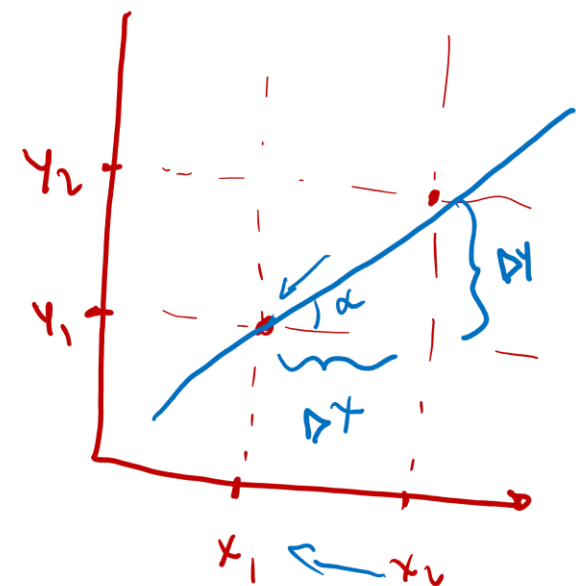


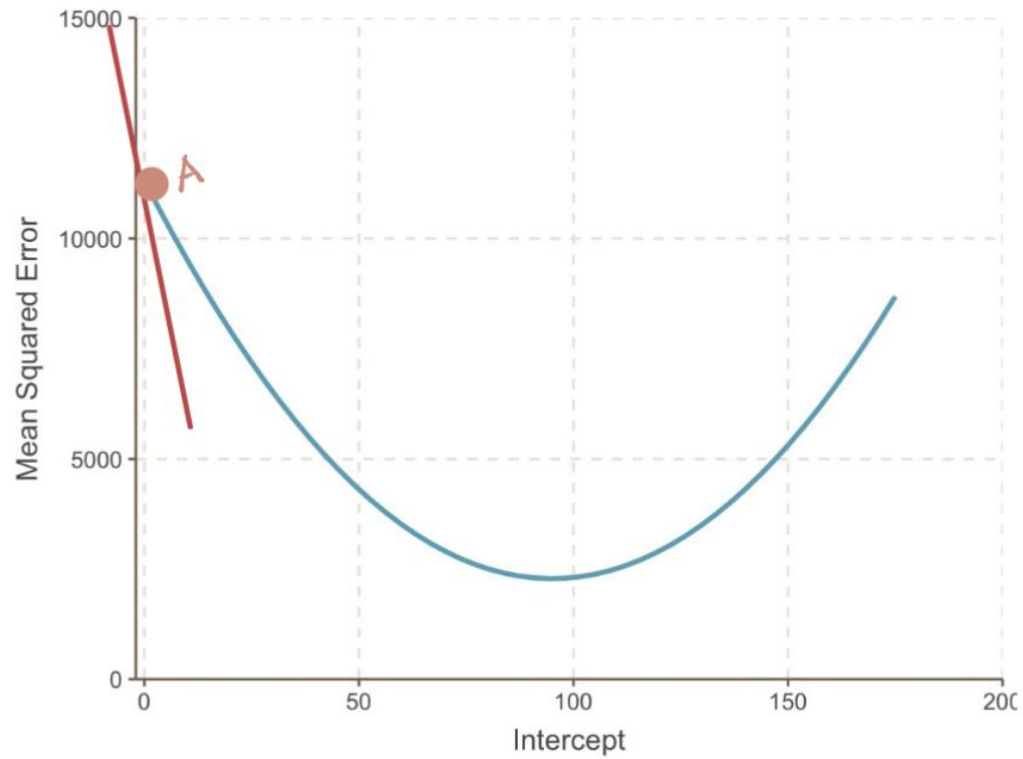
$$\begin{aligned} \text{MSE} &= \frac{1}{3} [(120 - (25 + 0.069 \times 1000))^2 + \\ &\quad (300 - (25 + 0.069 \times 2000))^2 + \\ &\quad (350 - (25 + 0.069 \times 4000))^2] \\ &= 7188 \end{aligned}$$





$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx}$$





① derivative is denoted by 'd'

② derivative with respect to the intercept

③ derivative at a specific value of the intercept

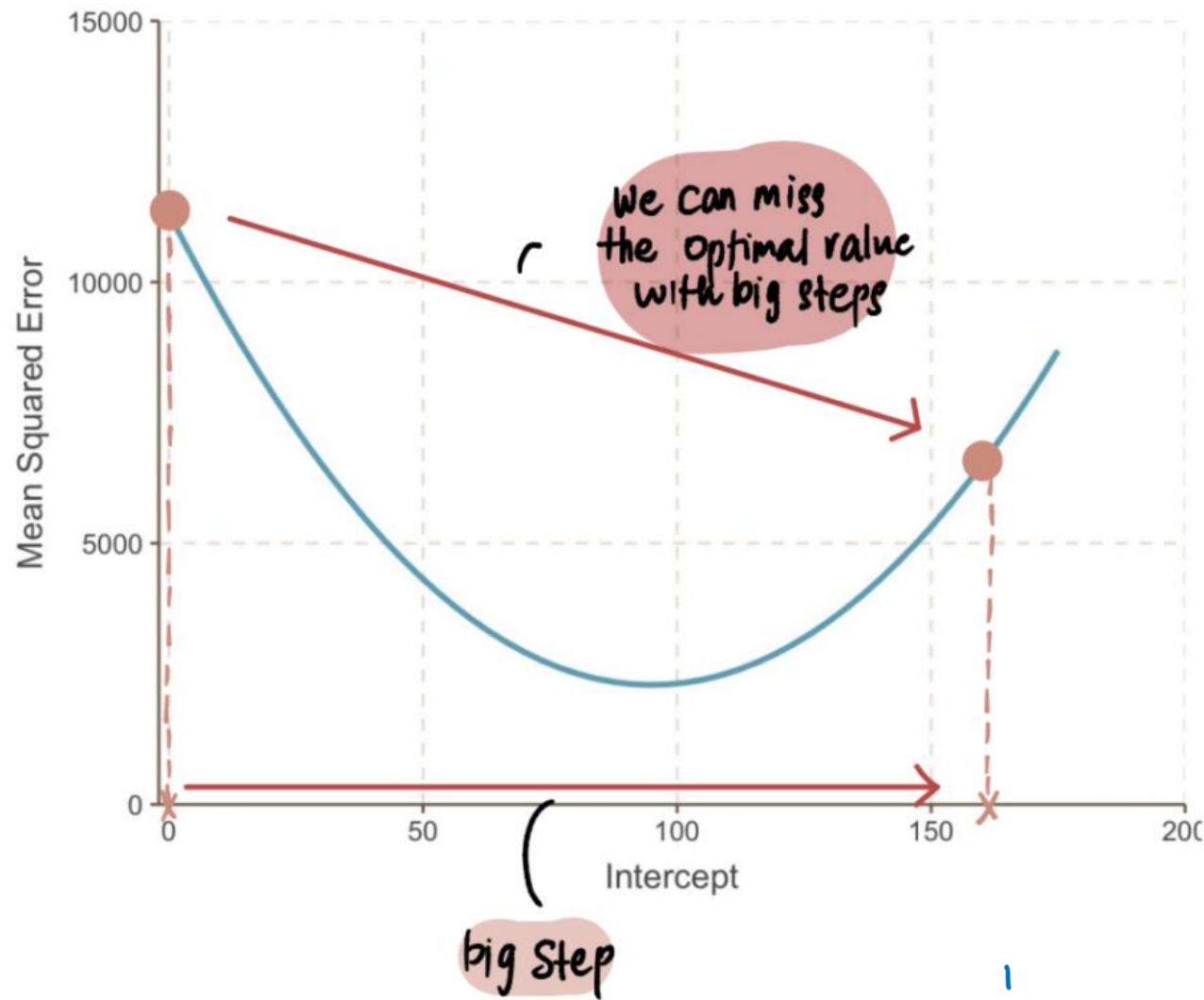
$\frac{d \text{ MSE}}{d \text{ intercept}} \bigg|_{\text{intercept}}$

$$\frac{dMSE}{d \text{ intercept}} = \frac{d}{d \text{ intercept}} \left[\frac{1}{3} \left[(120 - C \text{ intercept} + 0.069 \times 1000)^2 + (300 - C \text{ intercept} + 0.069 \times 2000)^2 + (350 - C \text{ intercept} + 0.069 \times 4000)^2 \right] \right]$$

$$= \frac{1}{3} \left[-2(120 - (\text{intercept} + 0.069 \times 1000)) - 2(300 - (\text{intercept} + 0.069 \times 2000)) - 2(350 - (\text{intercept} + 0.069 \times 4000)) \right]$$

$$\frac{dMSE}{d \text{ intercept}} \Big|_0 = \frac{1}{3} \left[-2(120 - (0 + 0.069 \times 1000)) - 2(300 - (0 + 0.069 \times 2000)) - 2(350 - (0 + 0.069 \times 4000)) \right]$$

$$= -190$$



$$\text{Step Size} = \text{Learning Rate} \times \text{gradient}$$

learning rate

gradient when intercept = 0

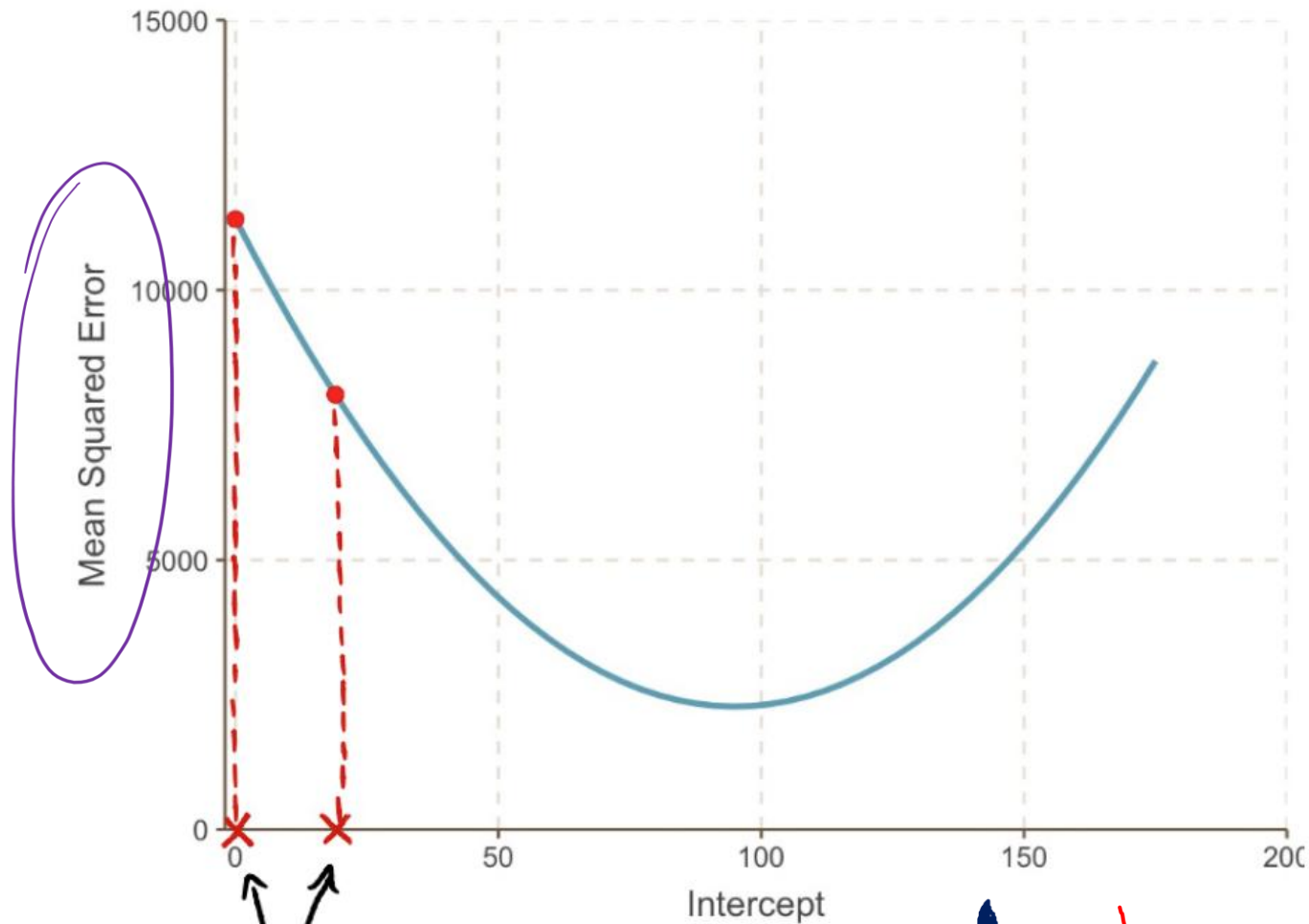
$$\text{Step Size} = 0.1 \times -190$$

$$= -19$$

$$\text{New Intercept} = \text{Old Intercept} - \text{Step Size}$$

$$= 0 - (-19)$$

$$= 19$$

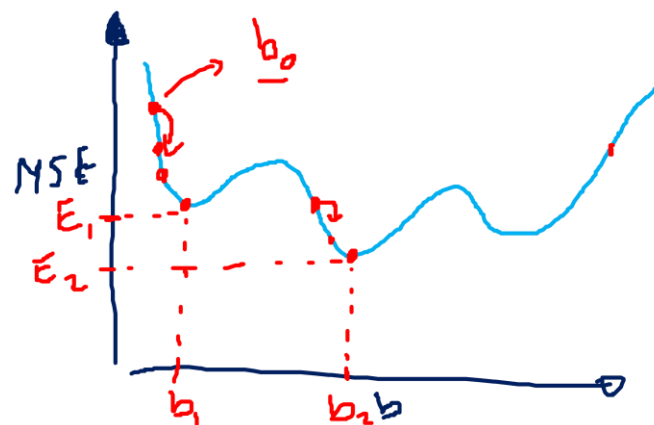


$$\text{New Intercept} = 0 - (-19) = 19$$

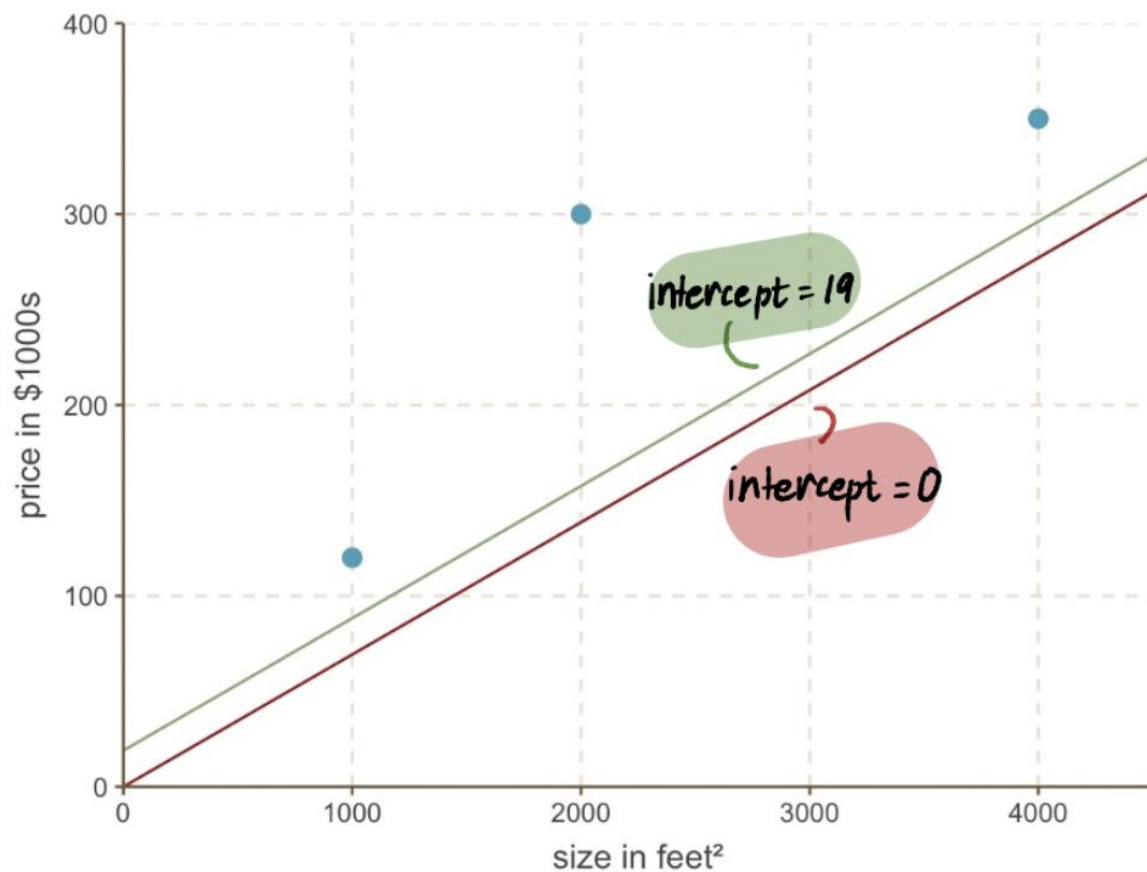


in one big Step,
we moved much
closer to our
optimal value

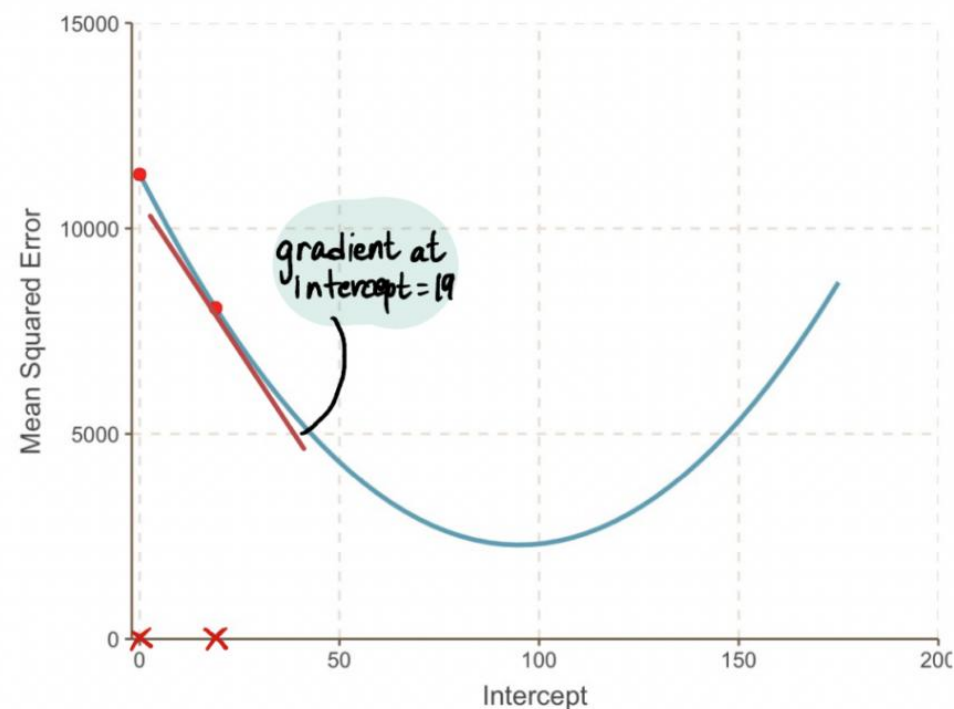
b

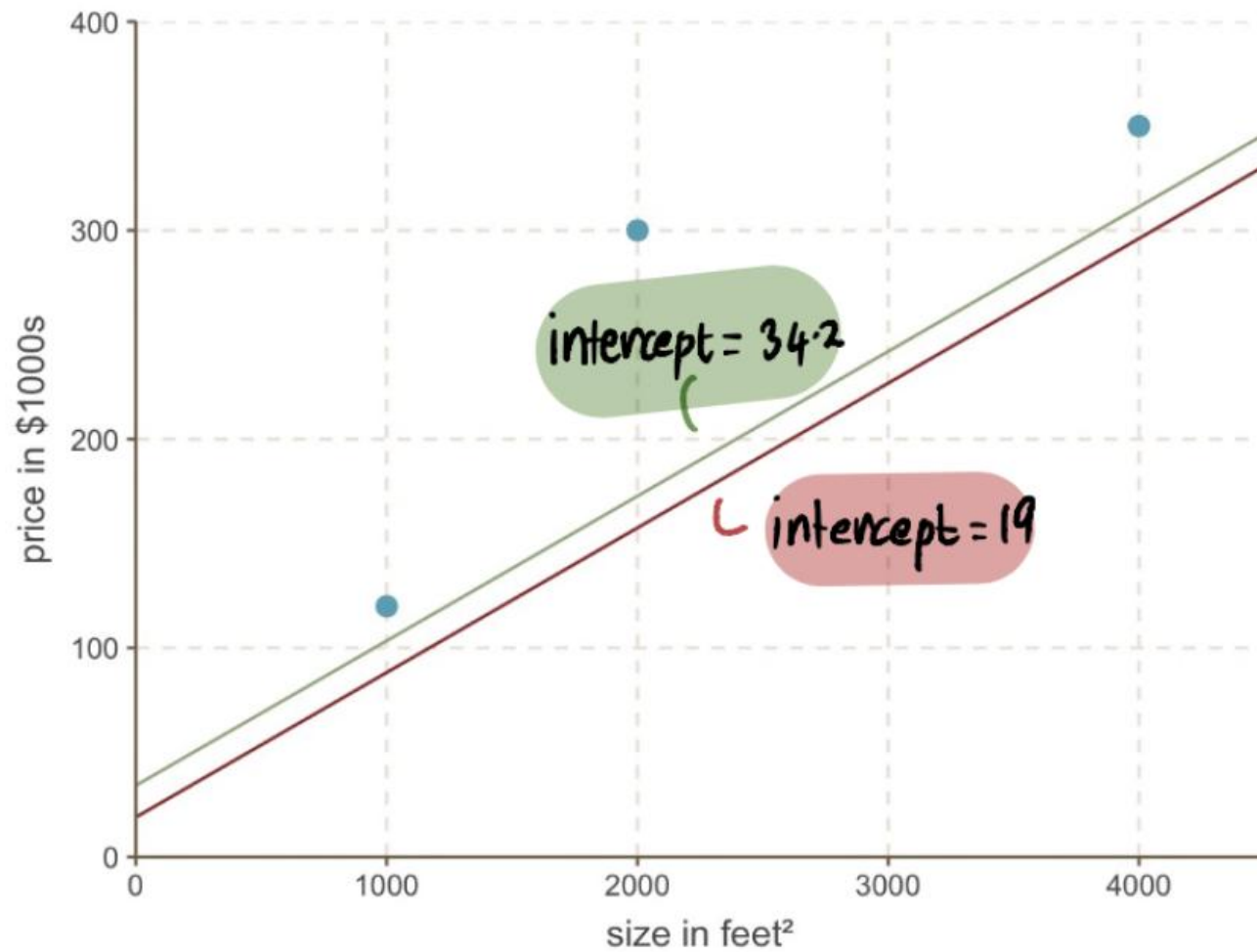


- $E_1 > E_2$
- 1.- Arg
 - 2.- w
 - 3.- $b = 0$
 - 4.- $\eta = 0.01$
- ✓



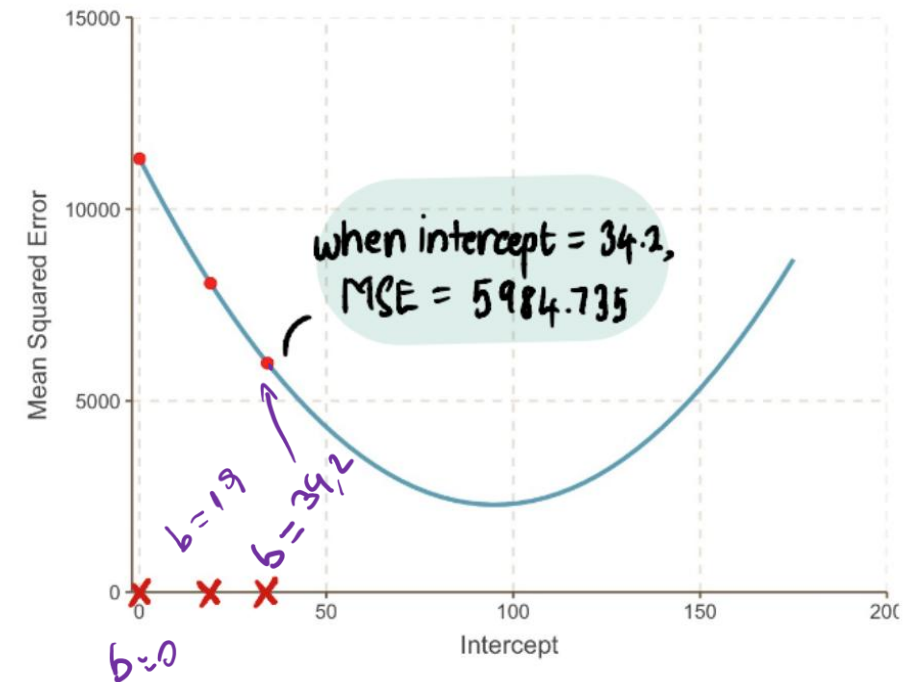
$$\begin{aligned} \frac{dMSE}{d \text{ intercept}} \Big|_{19} &= \frac{1}{3} [-2(120 - (19 + 0.069 \times 1000)) \\ &\quad - 2(300 - (19 + 0.069 \times 2000)) \\ &\quad - 2(350 - (19 + 0.069 \times 4000))] \\ &= -152 \end{aligned}$$

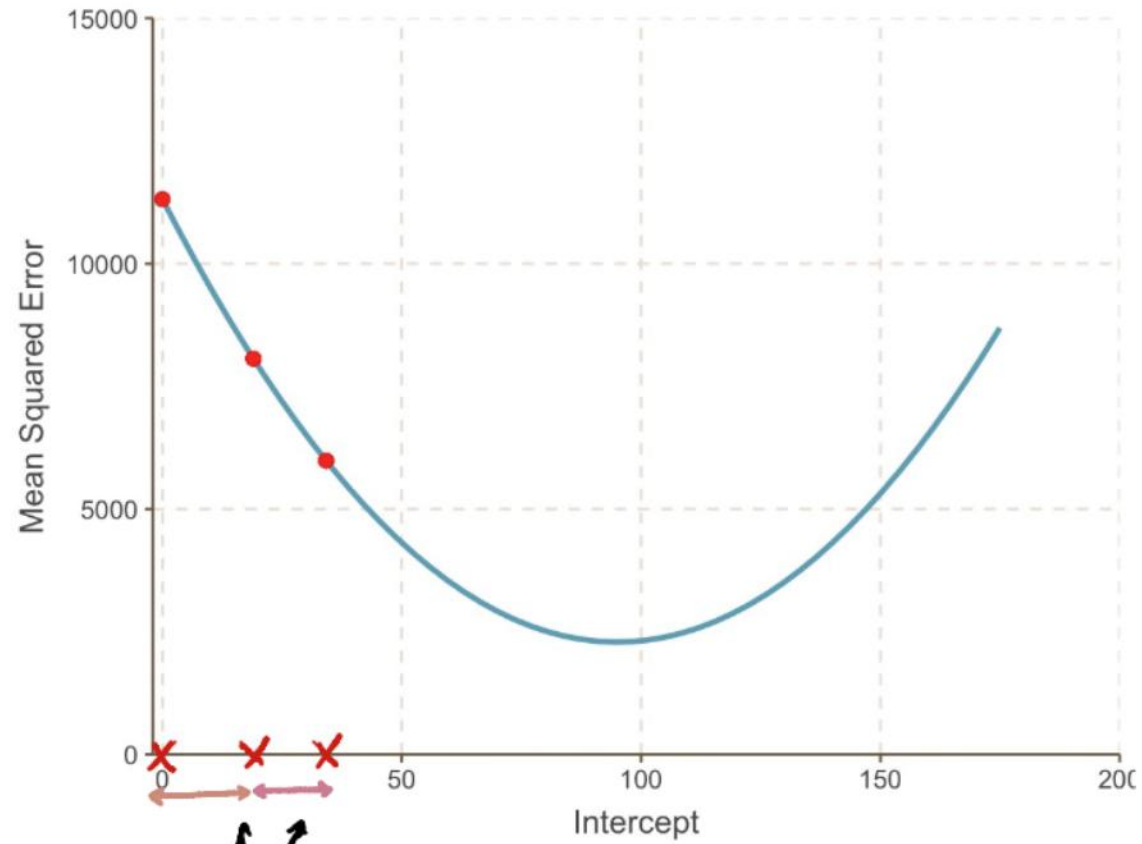




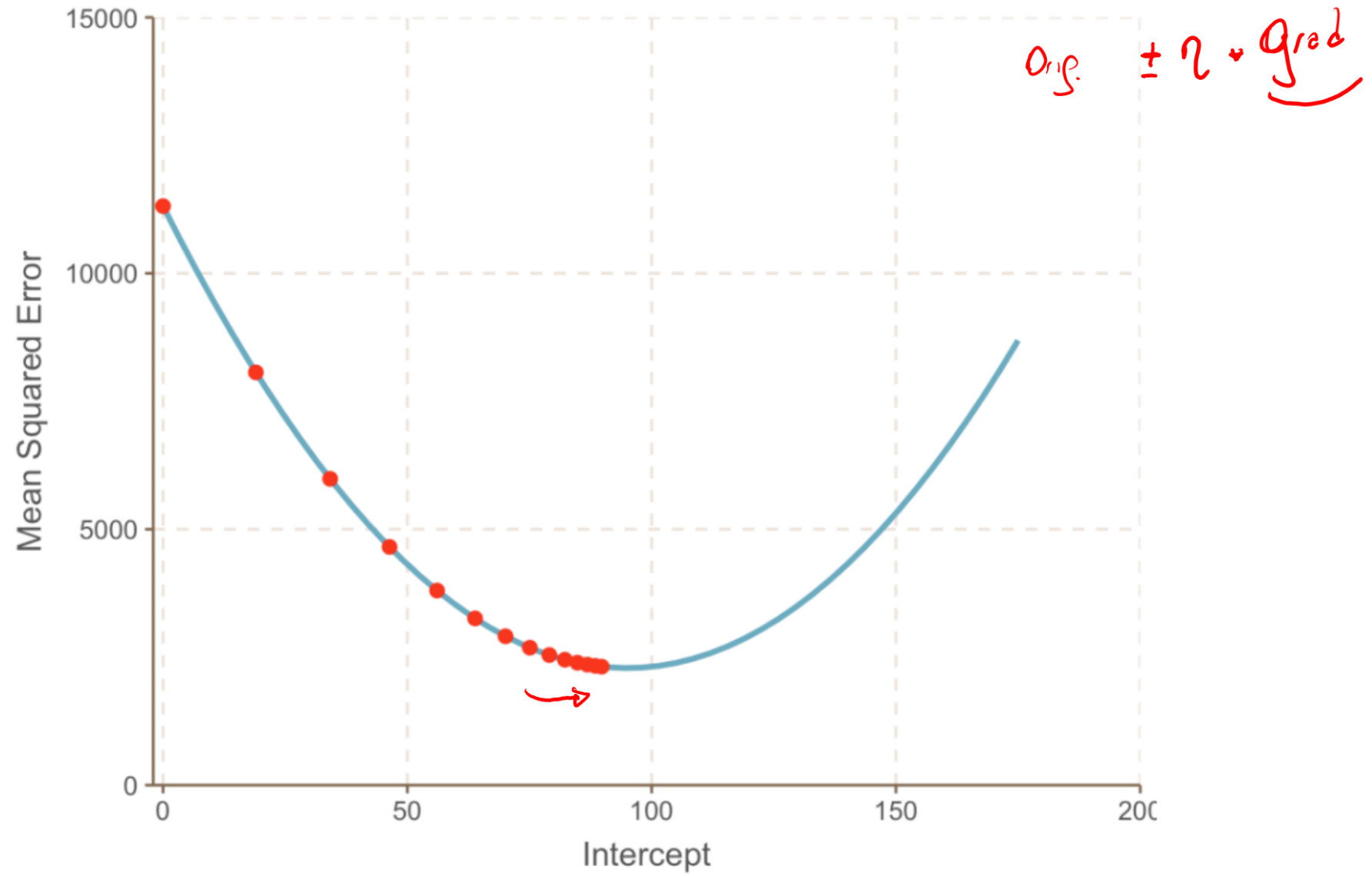
$$\begin{aligned}\text{Step Size} &= \text{Gradient} \times 0.1 \\ &= -152 \times 0.1 \\ &= \underline{-15.2}\end{aligned}$$

$$\begin{aligned}\text{New Intercept} &= \text{Old Intercept} - \text{Step Size} \\ &= 19 - (-15.2) \\ &= \underline{34.2}\end{aligned}$$





notice that
the first step was
slightly larger than
the first

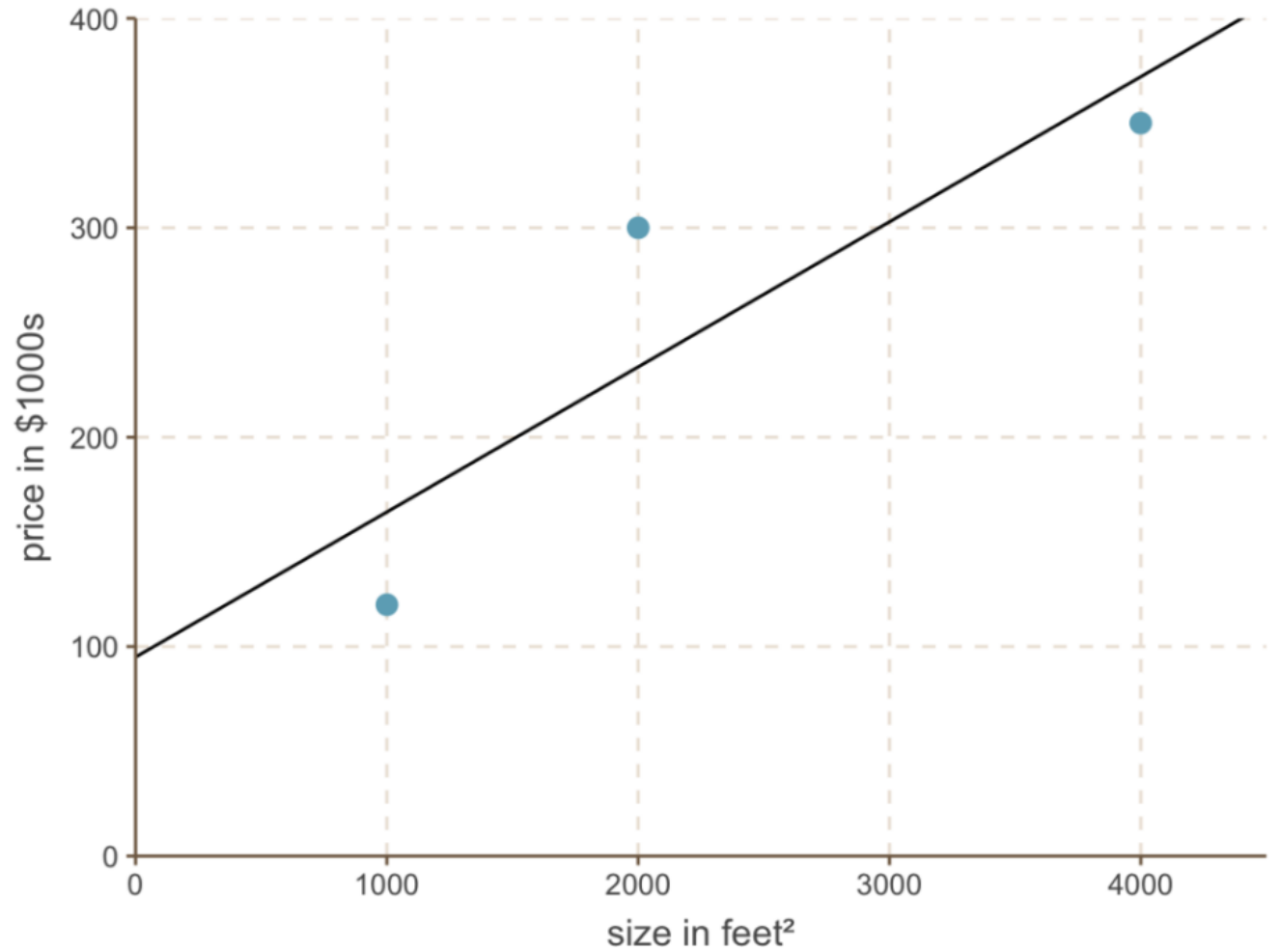


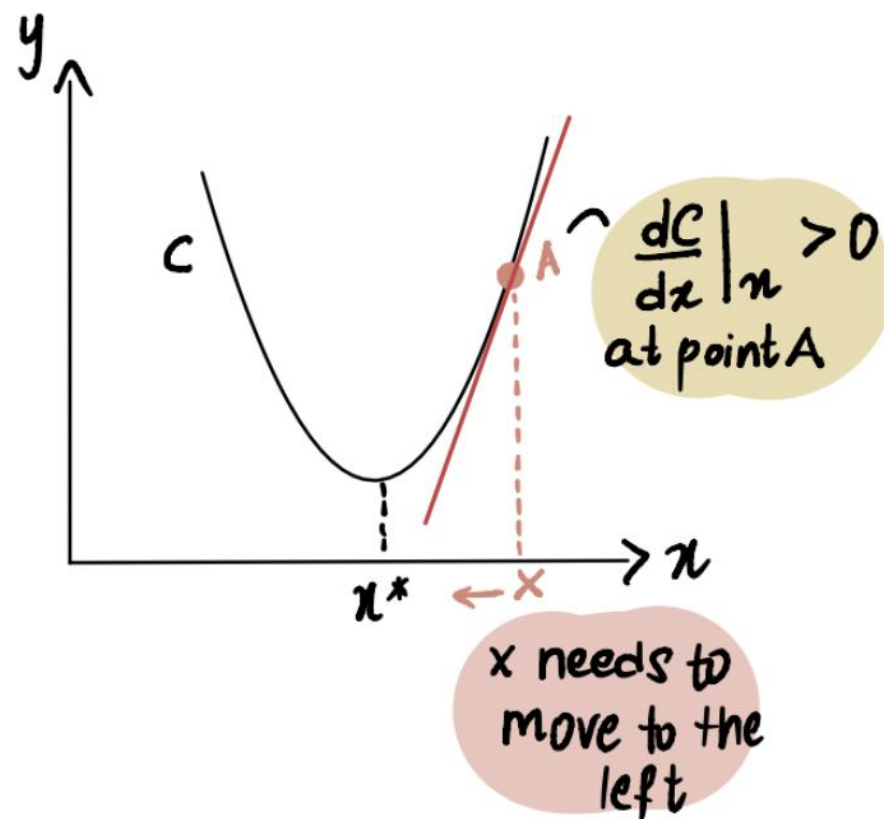
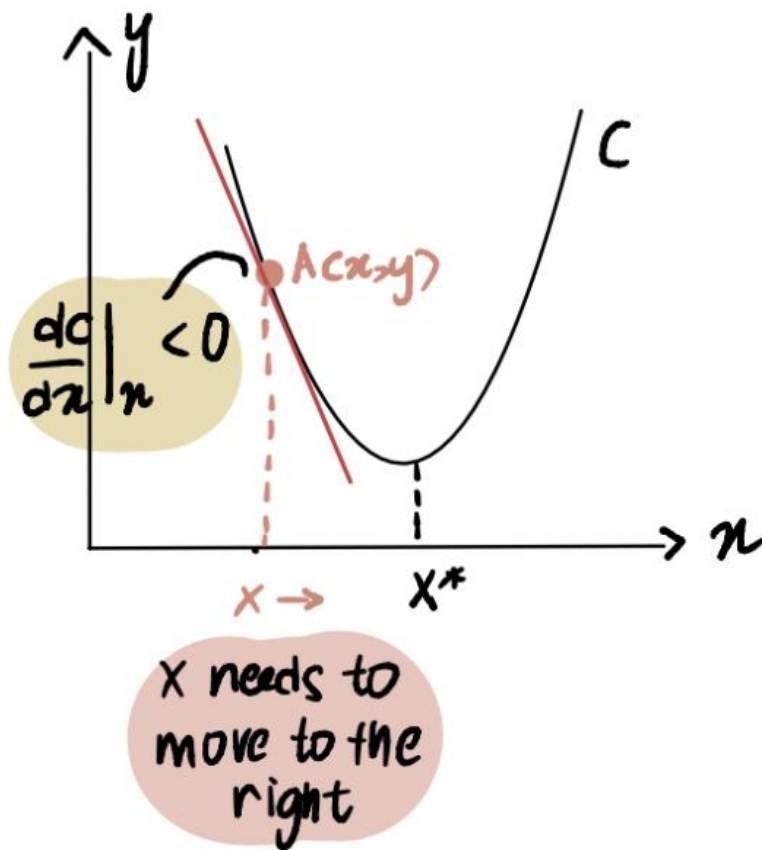
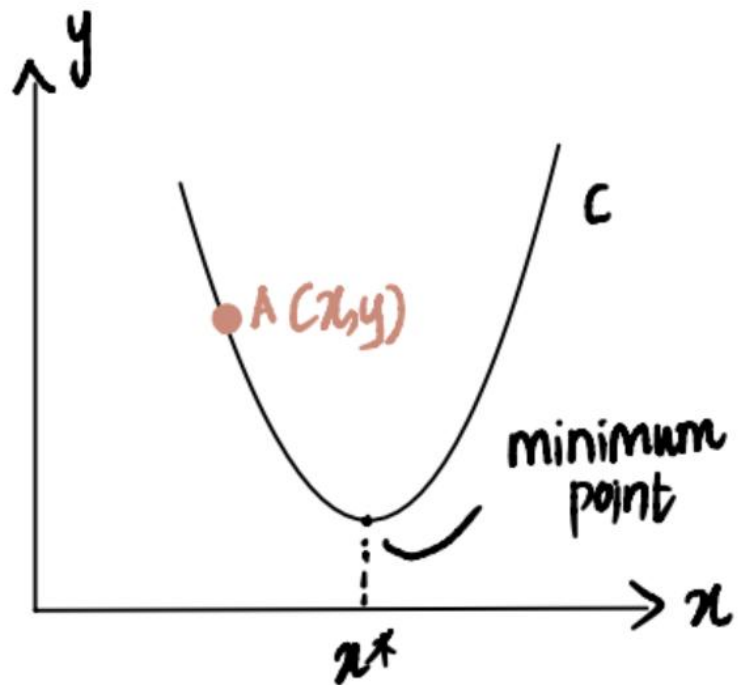
predicted house
price

$$\hat{y} = 95 + 0.069 \times \text{size}$$

optimal
intercept value
we found

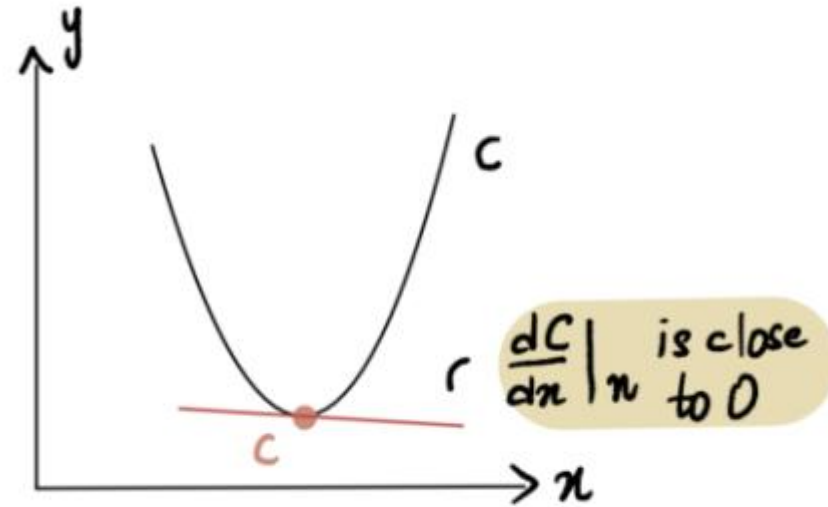
$$\begin{aligned}\hat{y} &= 95 + 0.069 \times 2400 \\ &= 260.6\end{aligned}$$

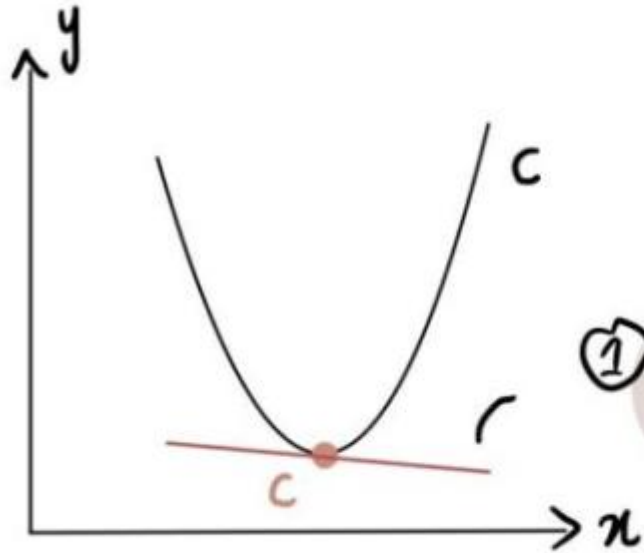




Step Size = Learning Rate $\times$ Gradient

Step size will be close to 0 when gradient is close to 0

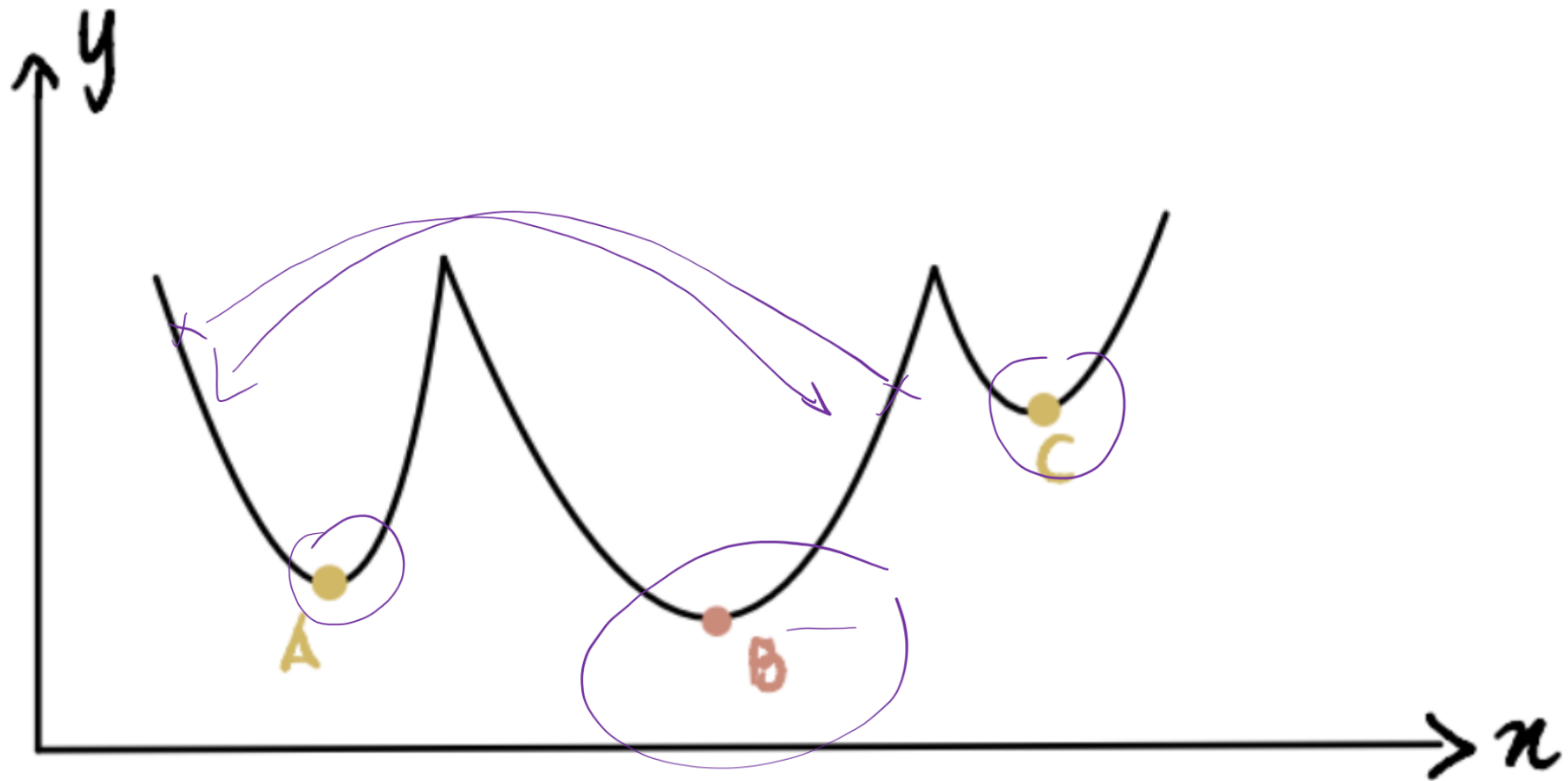


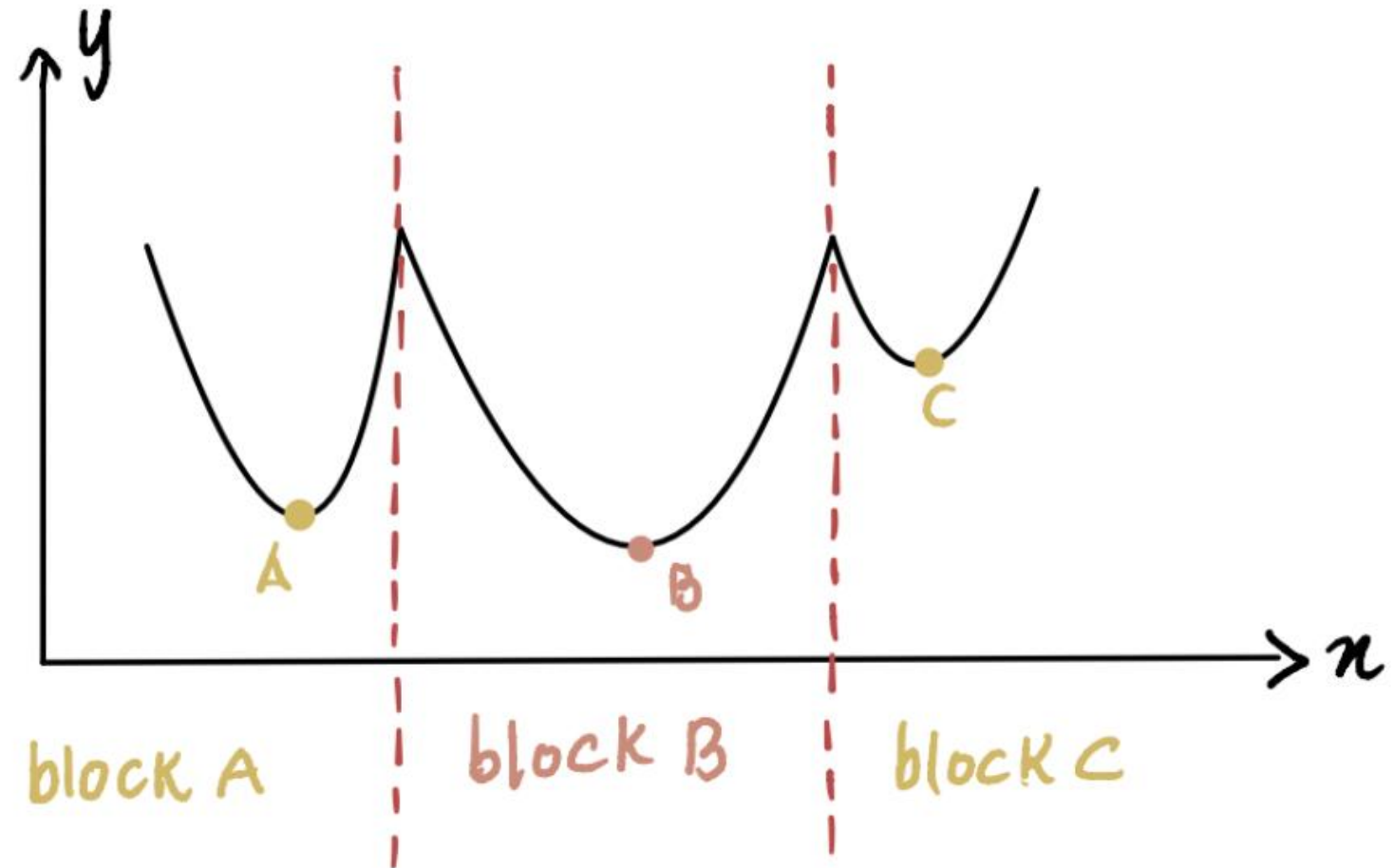


① if the gradient here = 0.009

② Step Size = 0.009×0.1
= 0.0009

③ $0.0009 < 0.001$,
So Gradient Descent
will stop





this symbol denotes partial derivative

$$\frac{\partial C(\pi_0, \pi_1)}{\partial \pi_0}, \quad \frac{\partial C(\pi_0, \pi_1)}{\partial \pi_1}$$

partial derivative with respect to π_0 partial derivative with respect to π_1

Step size for π_0

$$\pi_0 = \pi_0 - \alpha \times \frac{\partial C(\pi_0, \pi_1)}{\partial \pi_0}$$

$$\pi_1 = \pi_1 - \alpha \times \frac{\partial C(\pi_0, \pi_1)}{\partial \pi_1}$$

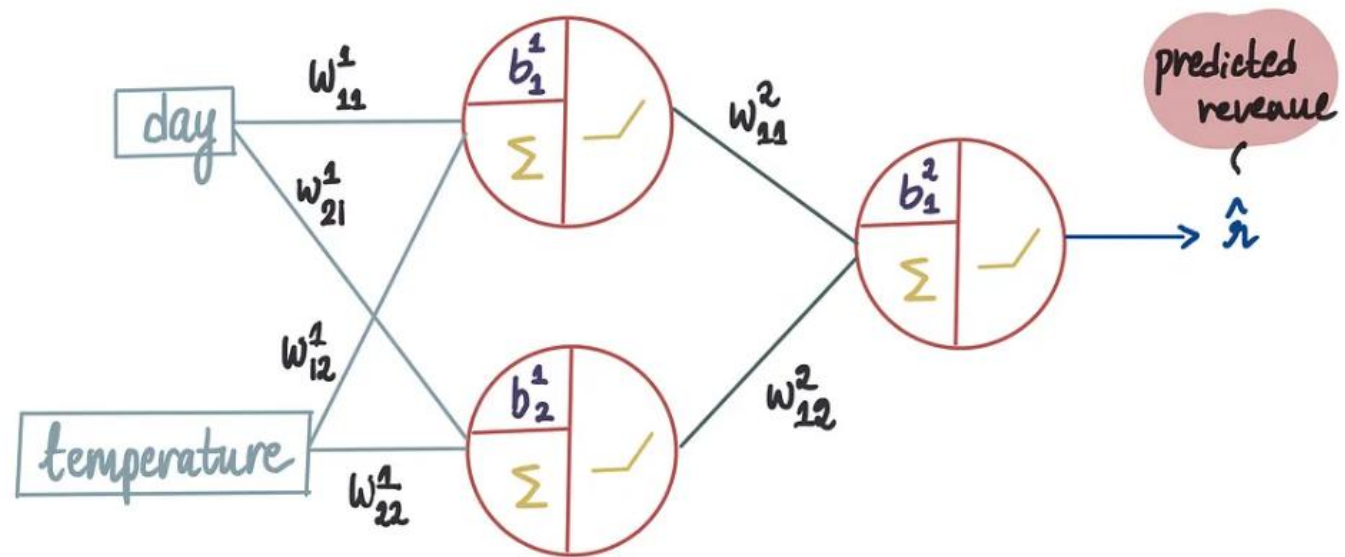
Step size for π_1

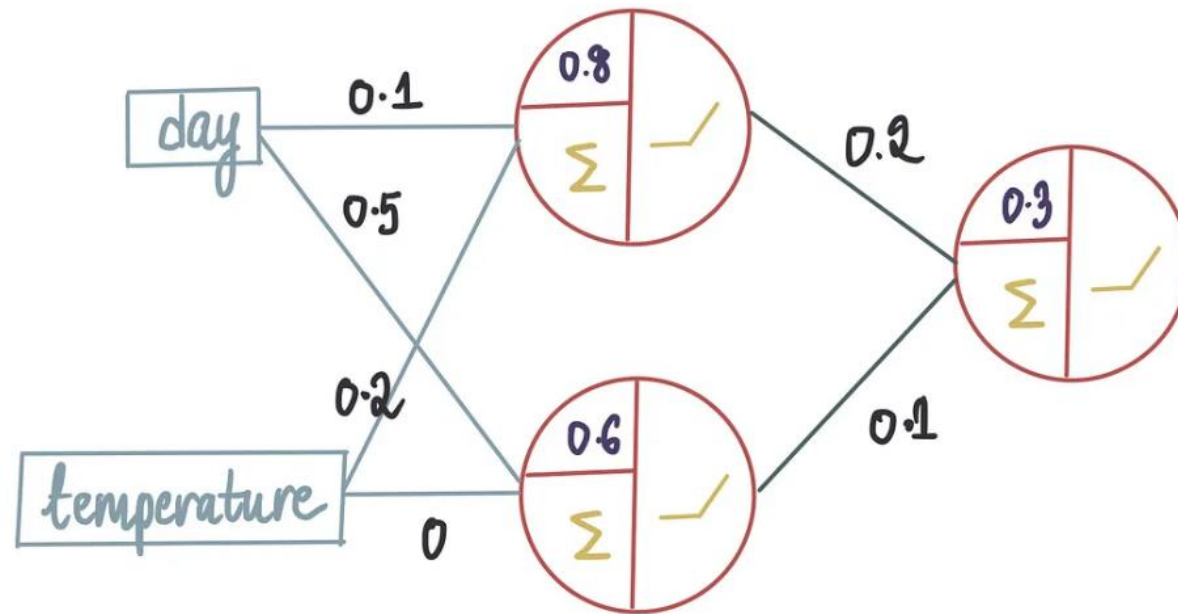
A Visual and Interactive Guide to the Basics of Neural
Networks – Jay Alammam – Visualizing machine learning one
concept at a time.

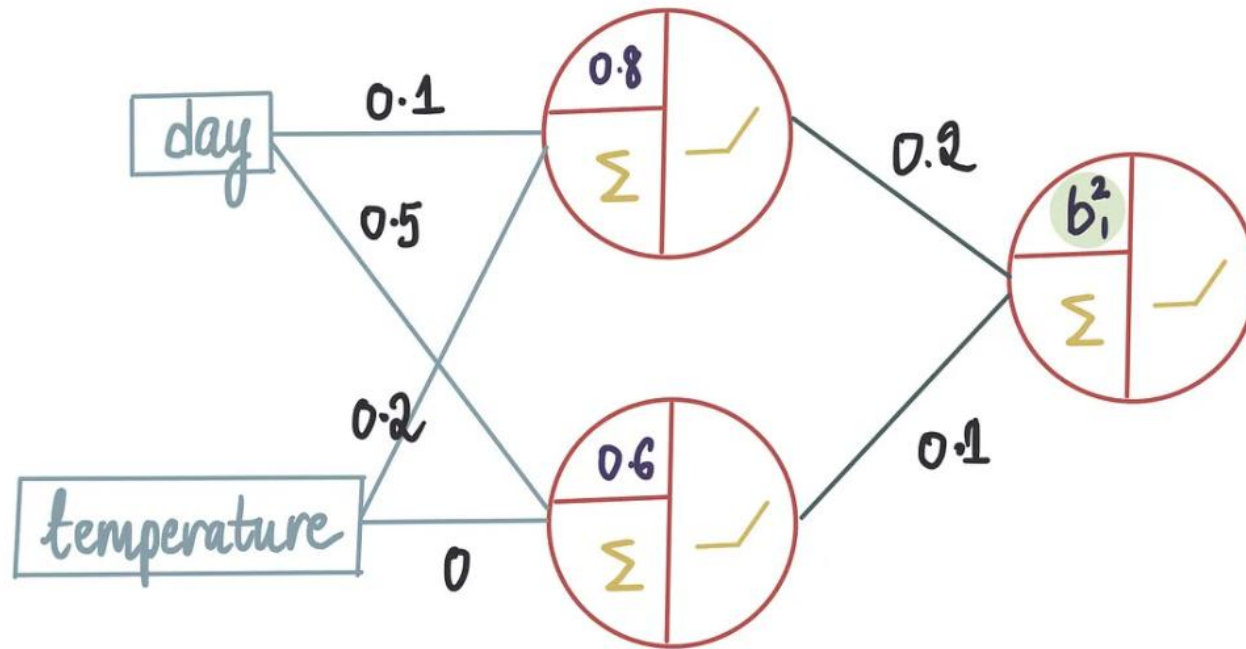
Introducción a Redes Neuronales

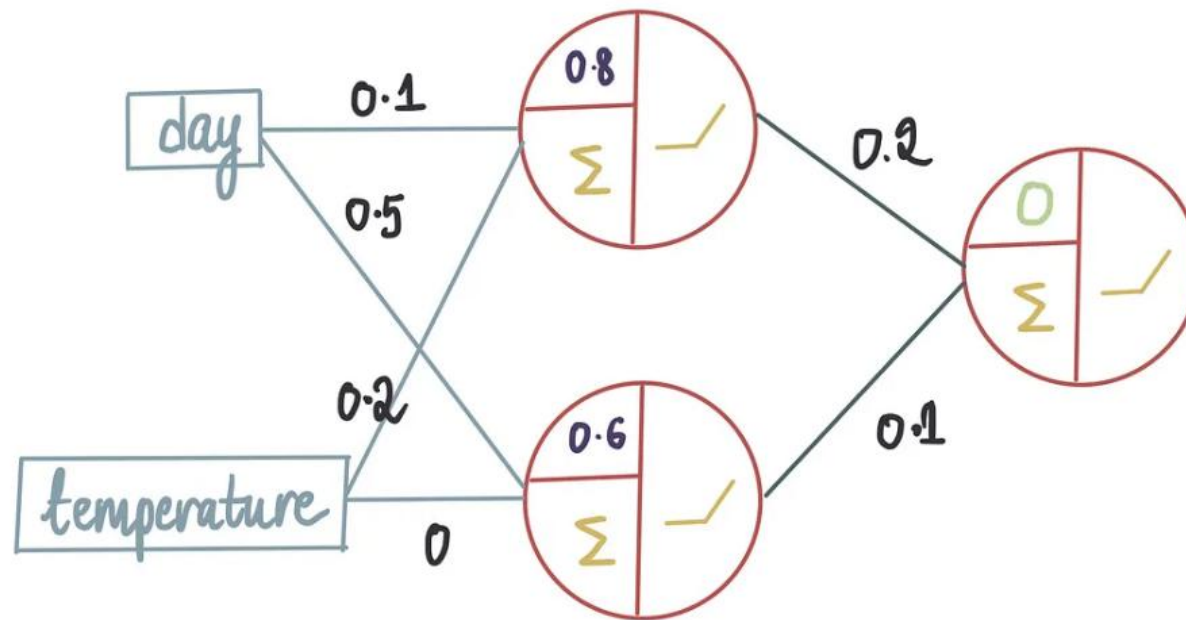
Back Propagation

	day	temperature (°C)	revenue (in 1000s)
<u>1</u> = Monday <u>2</u> = Tuesday ⋮ <u>7</u> = Sunday	2	22	1.51
	6	33	2.22
	1	20	1.37
	3	25	1.77
	2	24	1.64
	5	30	2.04
	7	35	2.42
	4	28	1.90
	3	26	1.75
	1	21	1.45

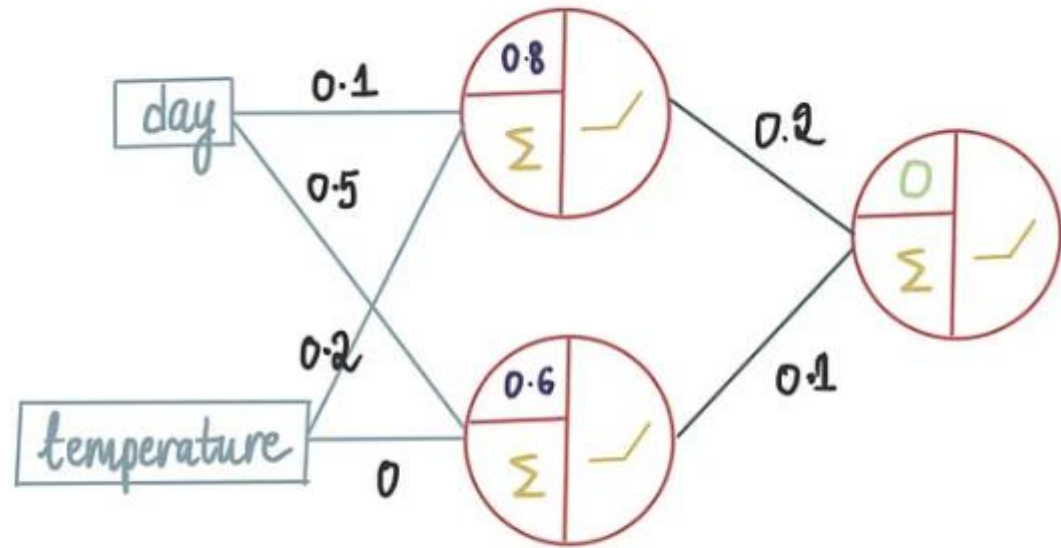
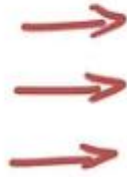








day	temperature (°C)
2	22
6	33
1	20
3	25
2	24
5	30
7	35
4	28
3	26
1	21

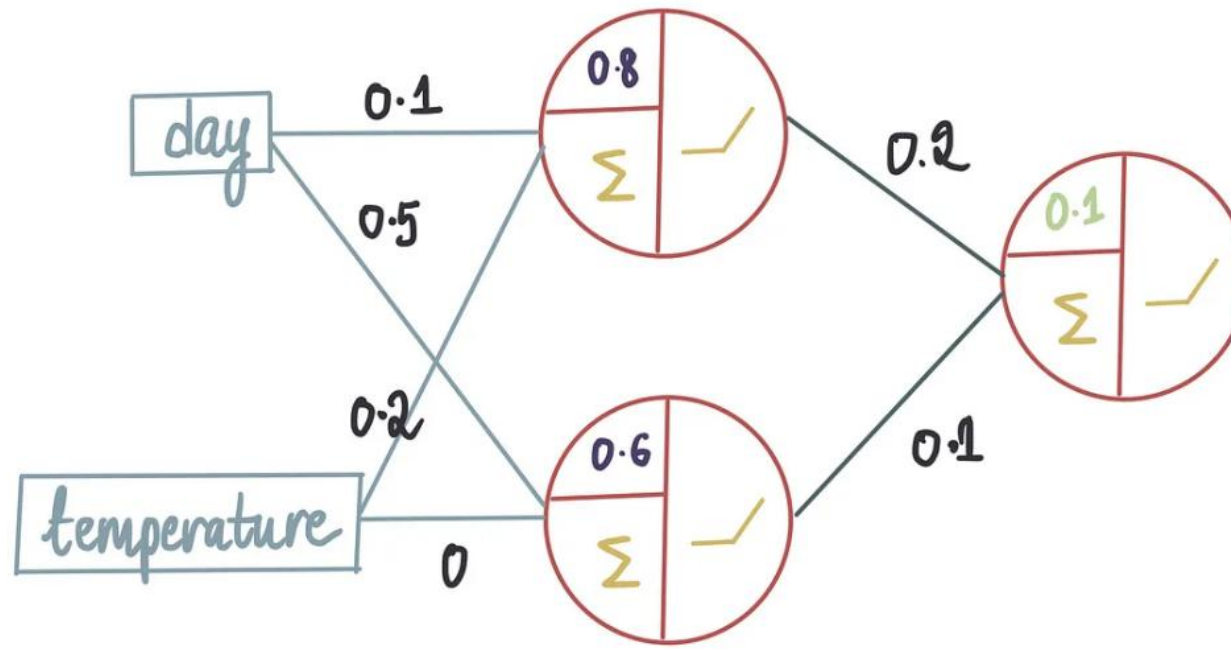


day	temperature (°C)	<u>revenue</u>	predicted revenue ($\hat{r}$) when <u>bias = 0</u>
2	22	1.51	1.24
6	33	2.22	1.96
1	20	1.37	1.09
3	25	1.77	1.43
2	24	1.64	1.32
5	30	2.04	1.77
7	35	2.42	2.11
4	28	1.90	1.62
3	26	1.15	1.47
1	21	1.45	1.23

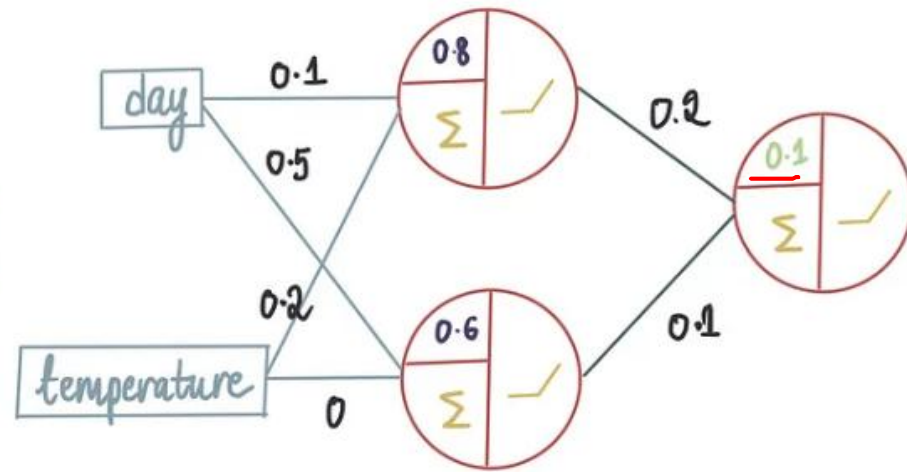
actual revenue predicted revenue

$$\underline{MSE} = \frac{1}{n} \sum (r - \hat{r})^2$$

$$\begin{aligned}\underline{MSE} &= \frac{1}{10} [(1.51 - 1.24)^2 + (2.22 - 1.96)^2 + \dots \\ &\quad + (1.75 - 1.47)^2 + (1.45 - 1.13)^2] \\ &= 0.0851\end{aligned}$$

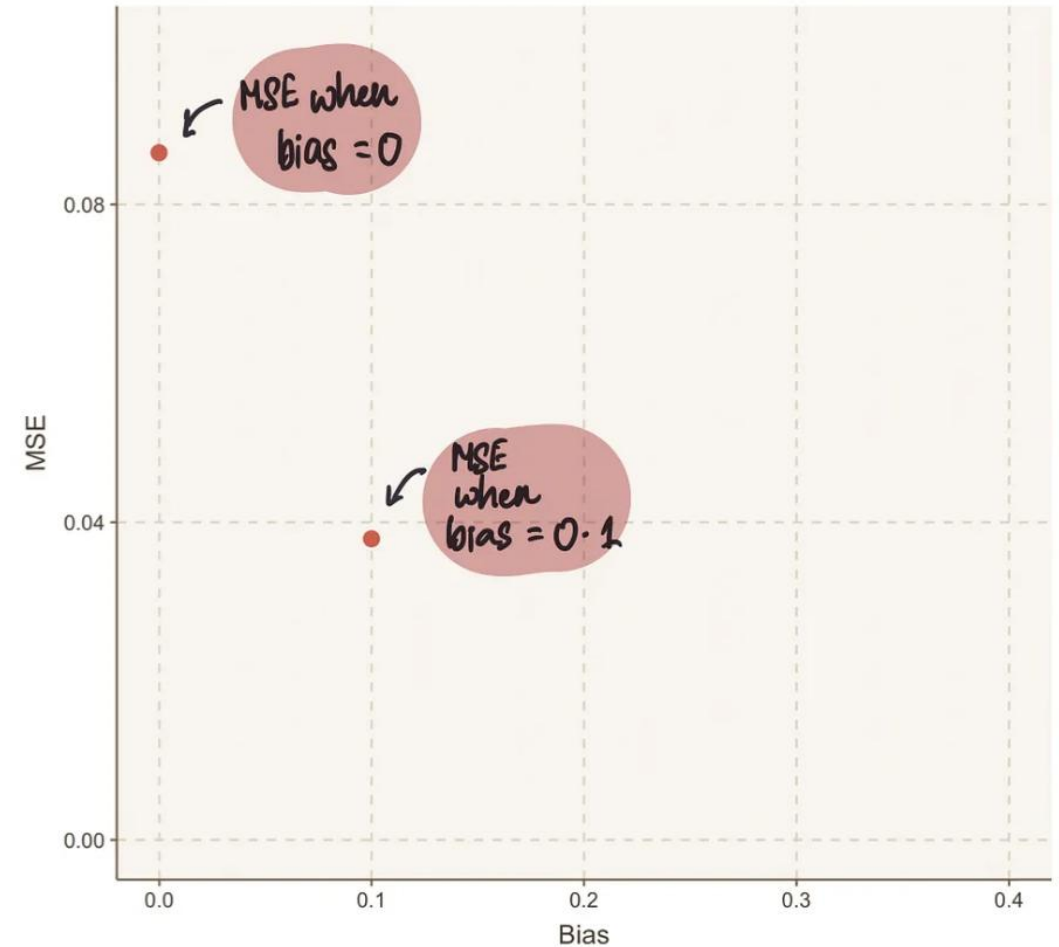


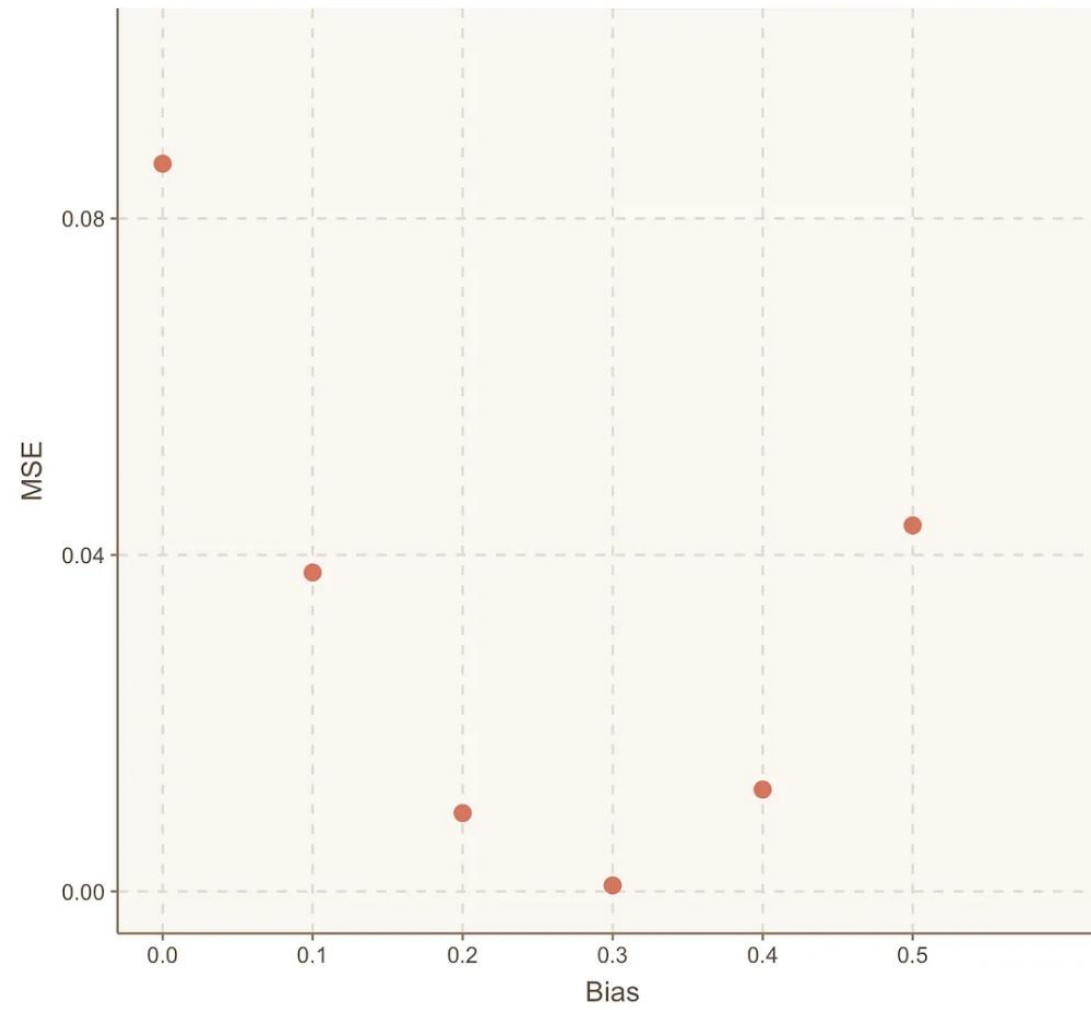
day	temperature (°C)
2	22
6	33
1	20
3	25
2	24
5	30
7	35
4	28
3	26
1	21

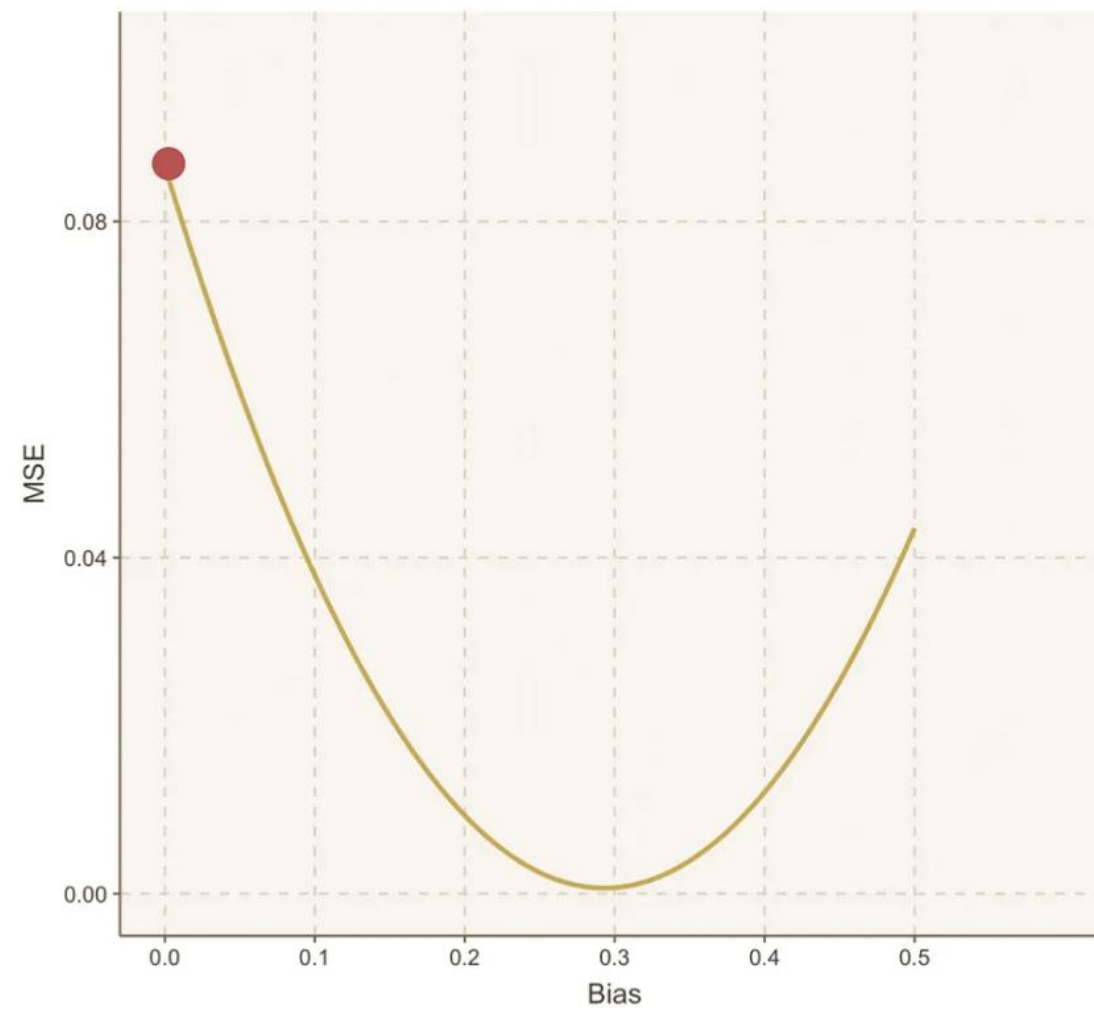


day	temperature (°C)	revenue	predicted revenue when <u>bias = 0.1</u>
2	22	<u>1.51</u>	<u>1.34</u>
6	33	2.22	2.06
1	20	1.37	1.19
3	25	1.77	1.53
2	24	1.64	1.42
5	30	2.04	1.87
7	35	2.42	2.21
4	28	1.90	1.72
3	26	1.75	1.57
1	21	<u>1.45</u>	1.23

$$\begin{aligned} \text{MSE} &= \frac{1}{10} [(1.51 - 1.34)^2 + (2.22 - 2.06)^2 + \dots \\ &\quad + (1.75 - 1.57)^2 + (1.45 - 1.23)^2] \\ &= 0.03791 \end{aligned}$$







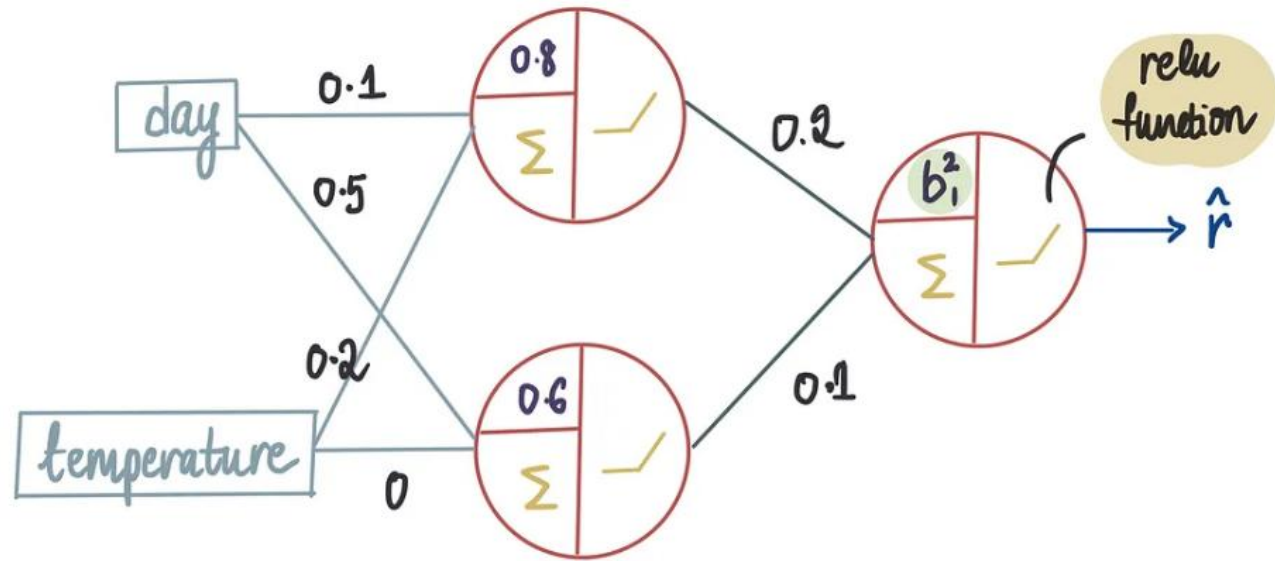
$$\text{Step size} = \text{learning rate} \times \frac{d\text{MSE}}{dbias} \Big|_{\text{bias}}$$

$$\text{Step size} = \text{learning rate} \times \left. \frac{d\text{MSE}}{dbias} \right|_{bias}$$

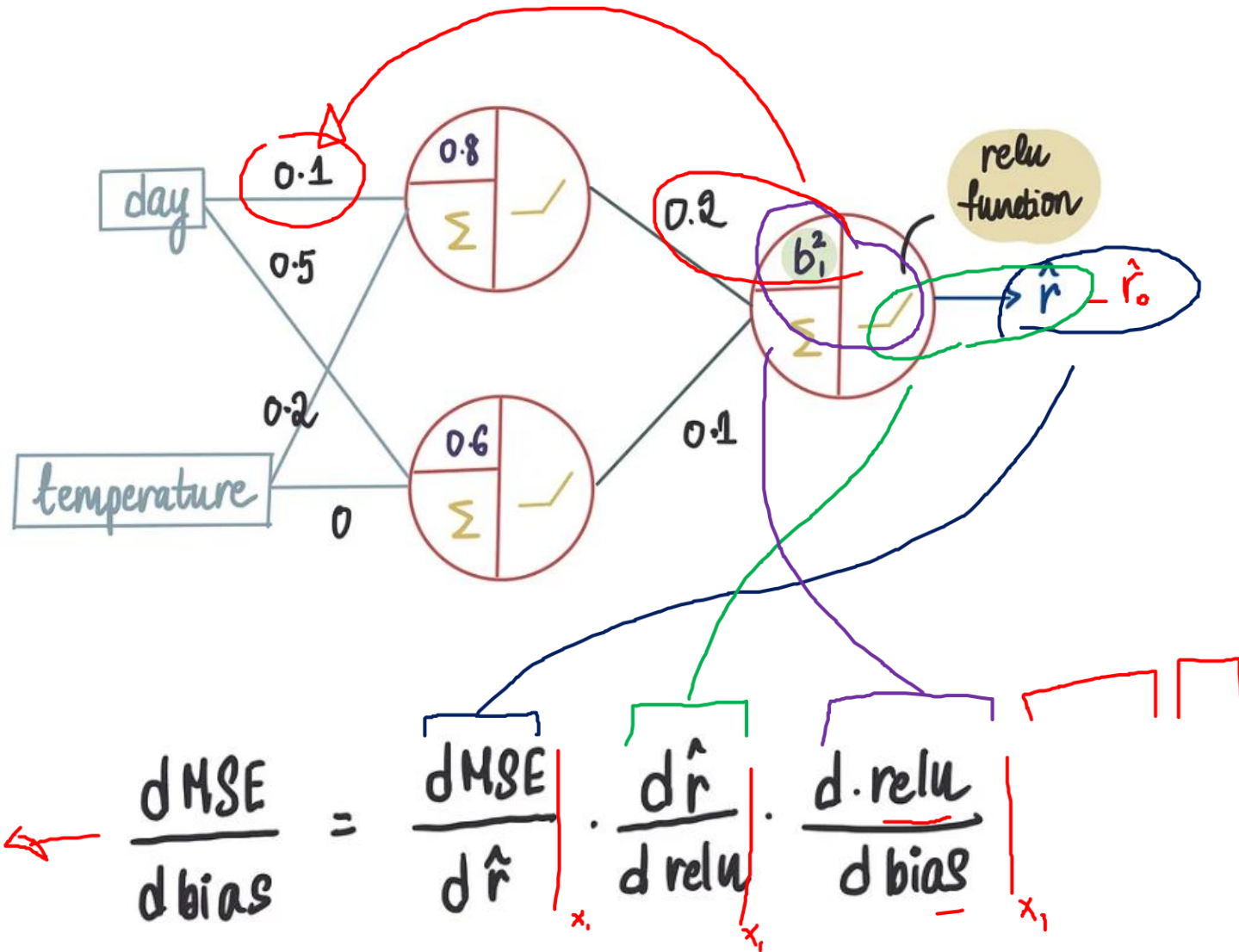
$$\text{MSE} = \frac{1}{n} \sum (r - \hat{r})^2$$

actual revenue

predicted revenue

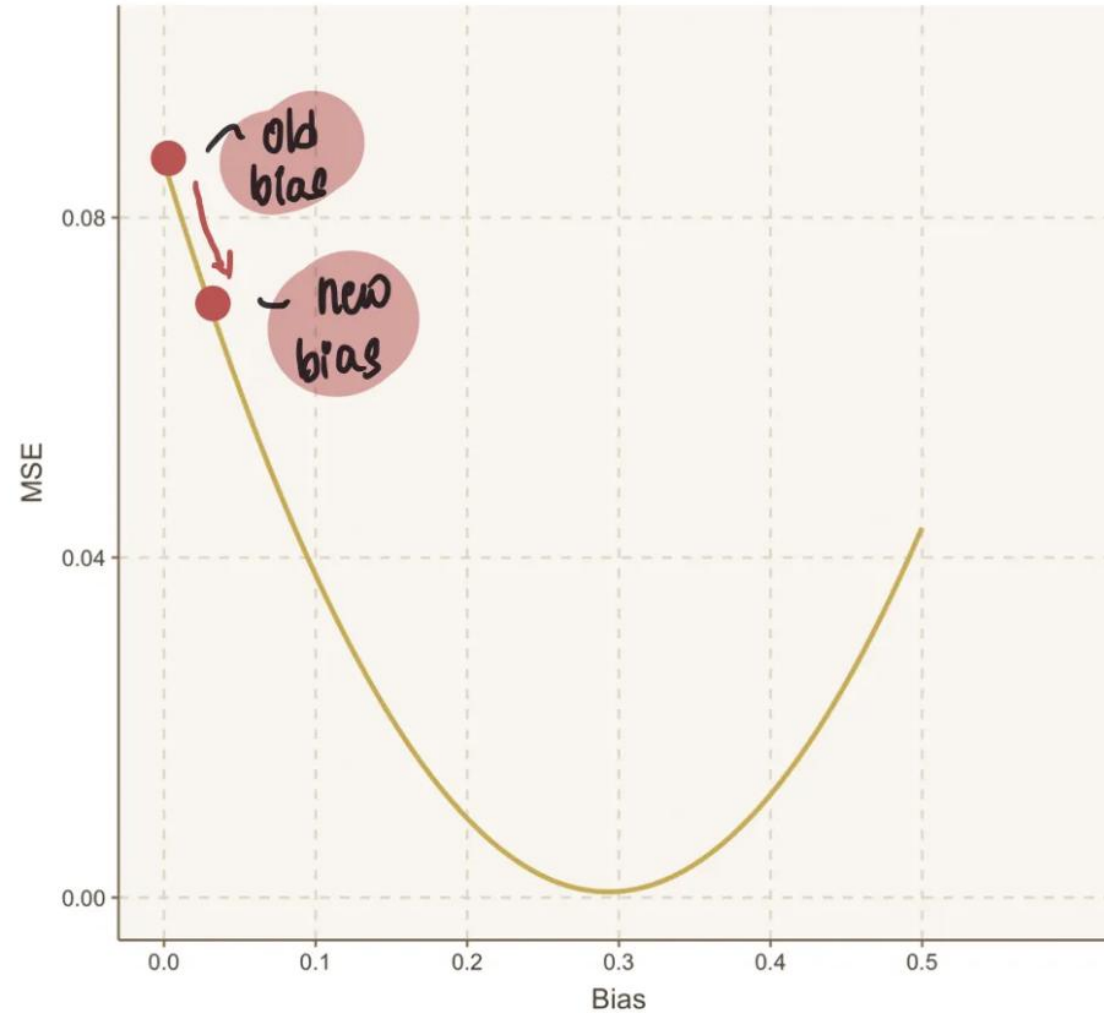


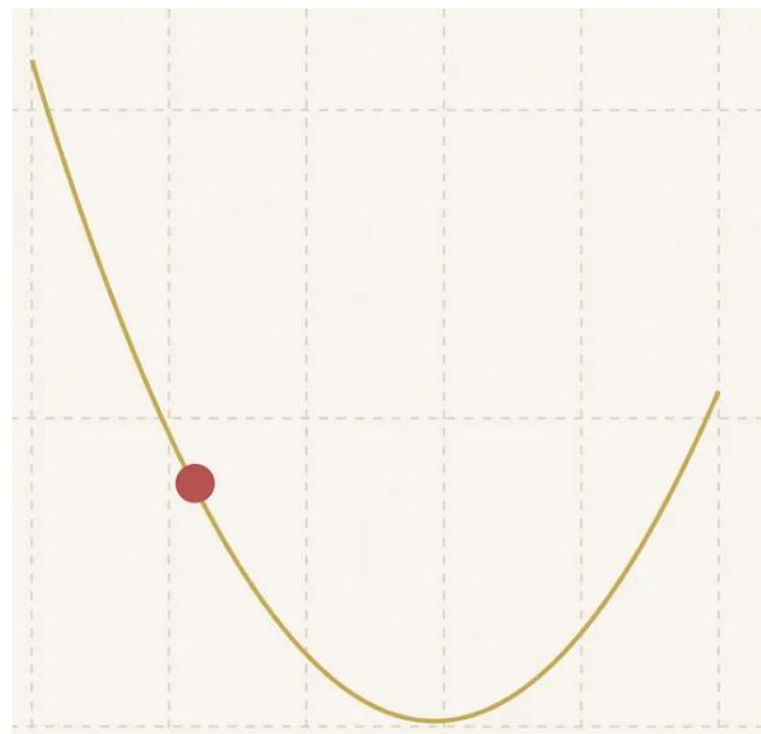
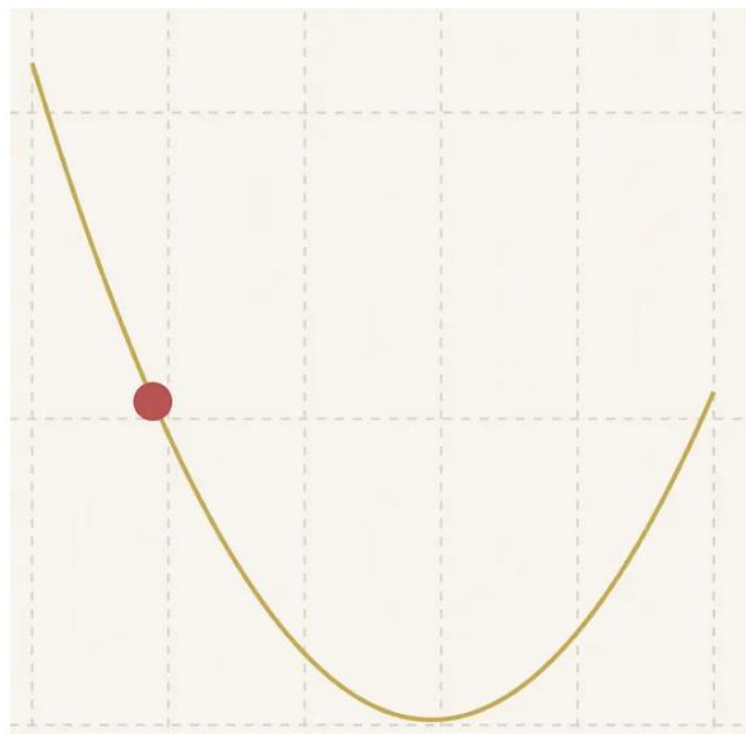
pytorch
↓
tensorflow



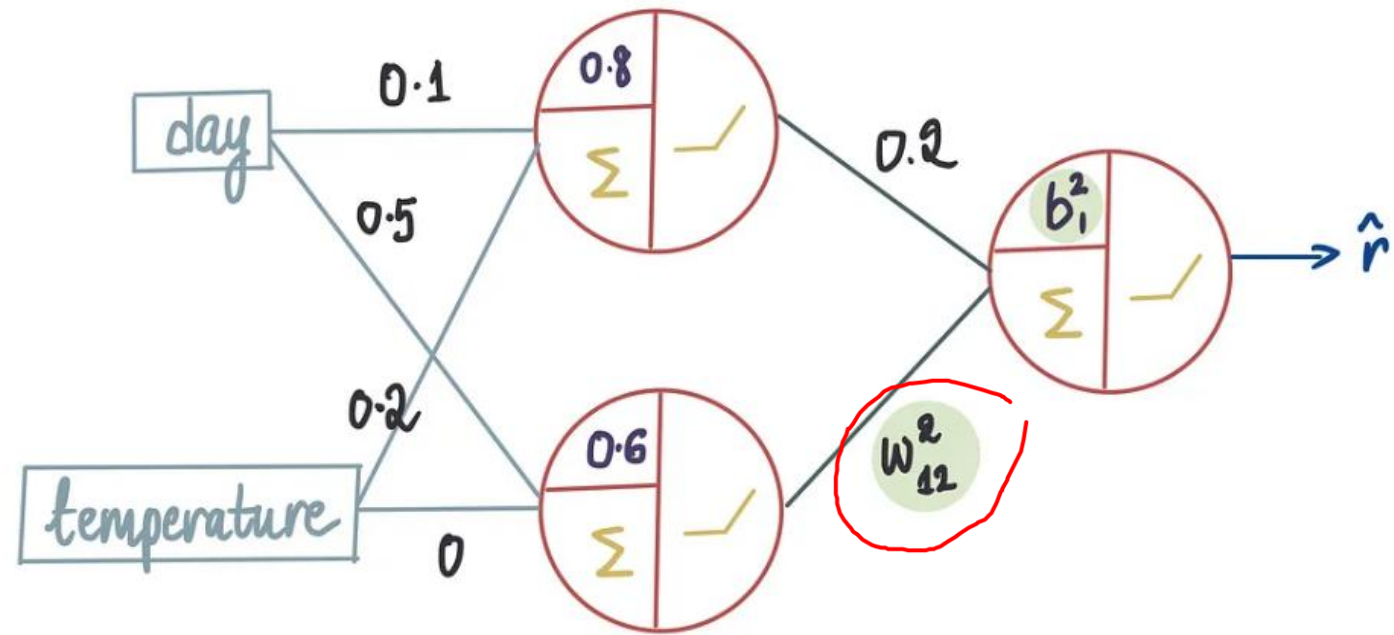
$$\begin{aligned}\text{new bias} &= \text{old bias} - \text{step size} \\ &= \text{old bias} - \text{learning rate} \times \left. \frac{d\text{MSE}}{dbias} \right|_{\text{old bias}}\end{aligned}$$

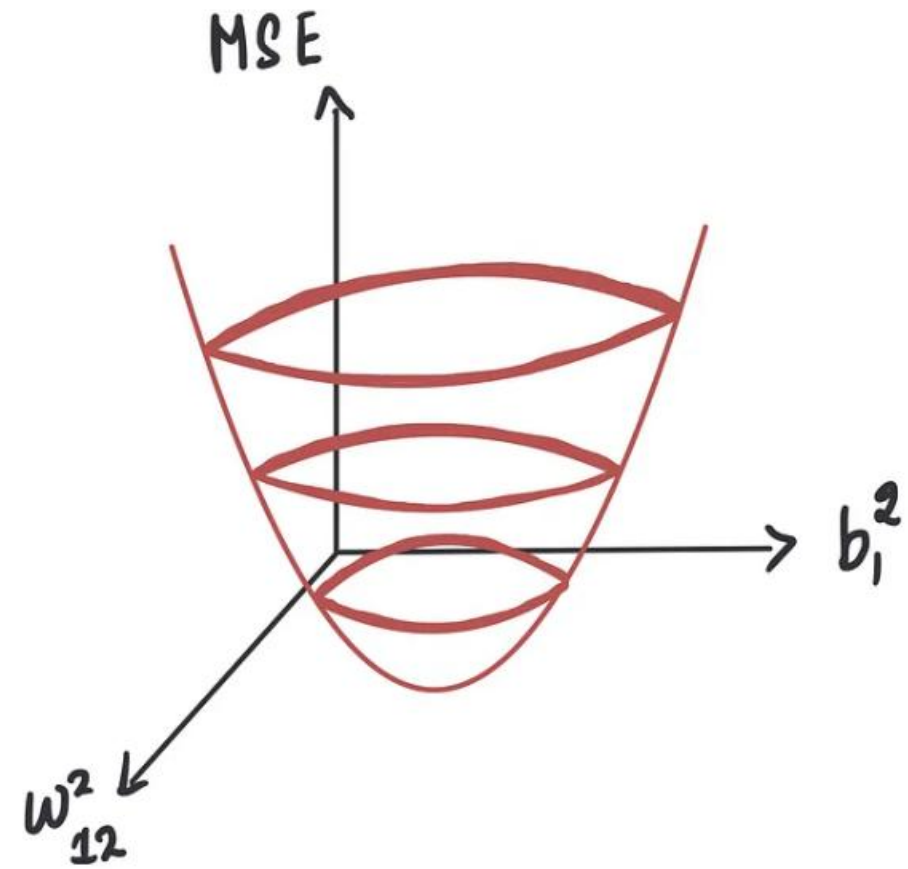
$$\begin{aligned}\text{new bias} &= \text{old bias} - \text{step size} \\ &= \text{old bias} - \text{learning rate} \times \left. \frac{d\text{MSE}}{dbias} \right|_{\text{old bias}}\end{aligned}$$











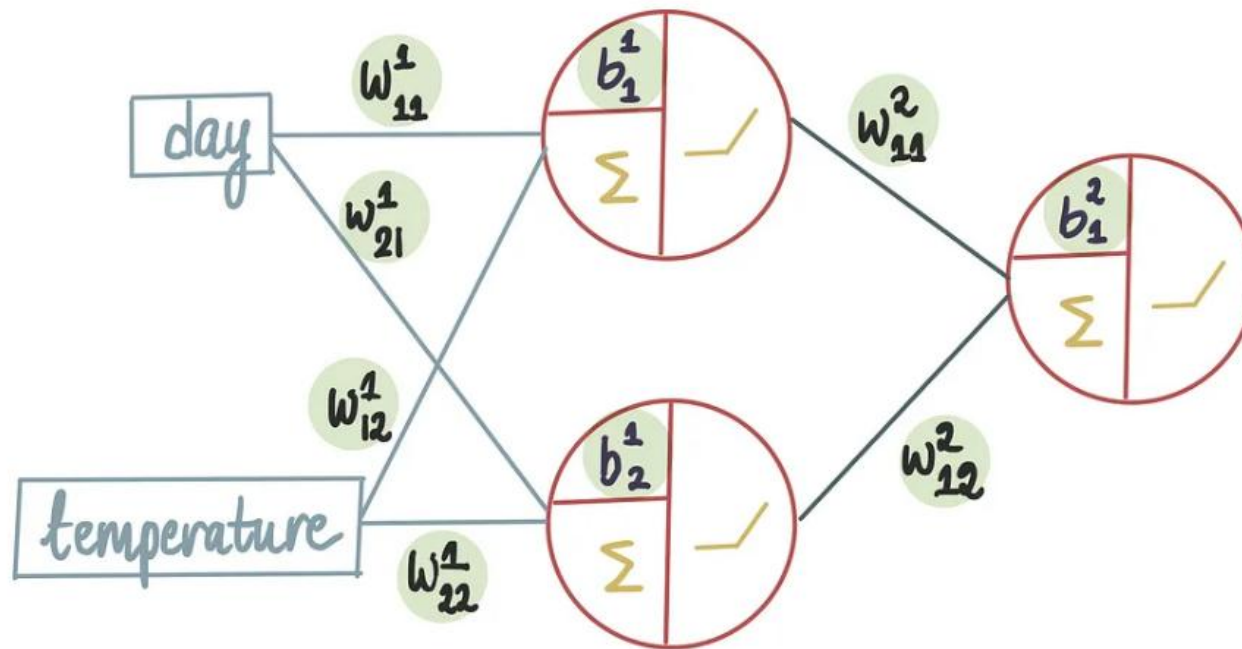
$$\text{new } b_1^2 \text{ step size} = (\text{learning rate}) \cdot \frac{\partial \text{MSE}}{\partial b_1^2}$$

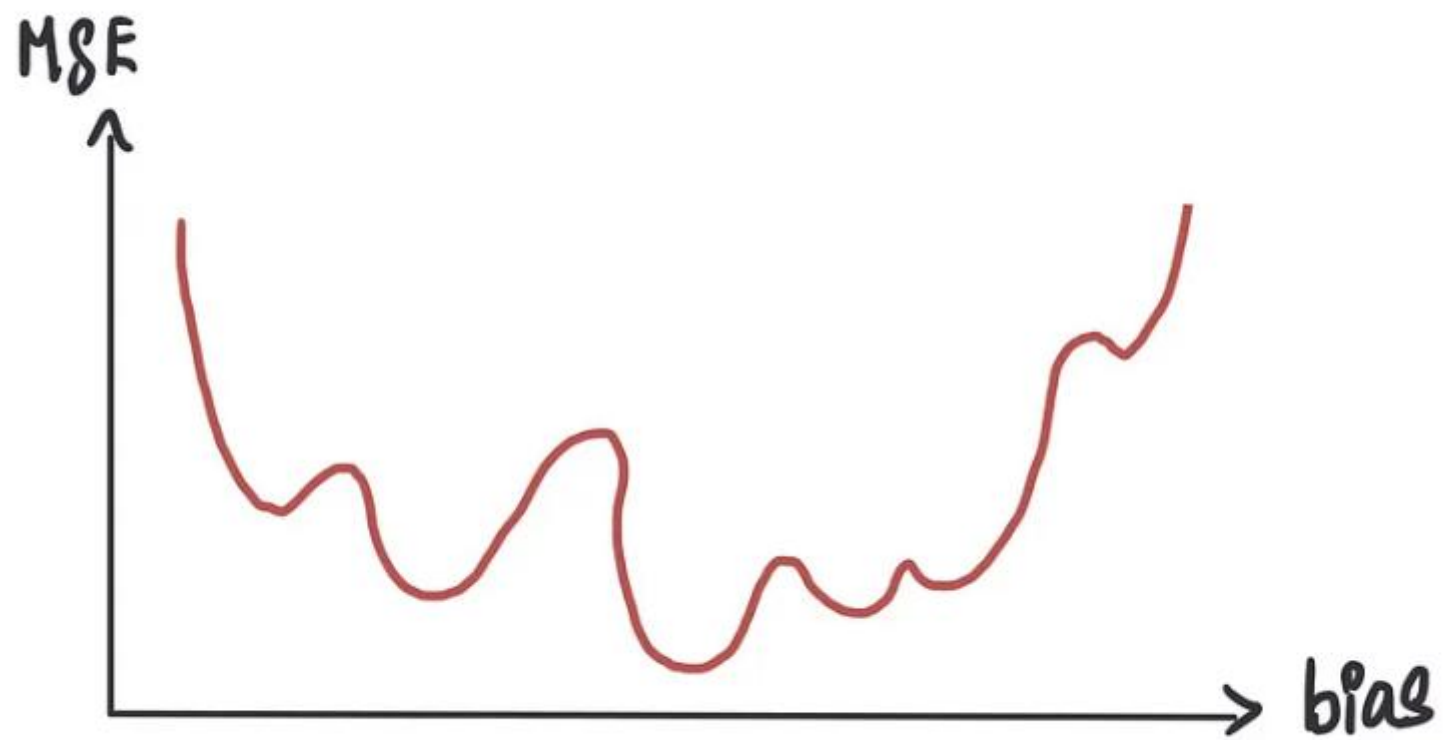
$$\text{new } w_{12}^2 \text{ step size} = (\text{learning rate}) \cdot \frac{\partial \text{MSE}}{\partial w_{12}^2}$$

$$\frac{\partial \text{MSE}}{\partial b_1^2} = \frac{\partial \text{MSE}}{\partial \hat{r}} \cdot \frac{\partial \hat{r}}{\partial \text{relu}} \cdot \frac{\partial \text{relu}}{\partial b_1^2}$$

$$\frac{\partial \text{MSE}}{\partial w_{12}^2} = \frac{\partial \text{MSE}}{\partial \hat{r}} \cdot \frac{\partial \hat{r}}{\partial \text{relu}} \cdot \frac{\partial \text{relu}}{\partial w_{12}^2}$$

$$\begin{aligned} \text{new } b_1^2 &= \text{old } b_1^2 - (\text{learning rate}) \cdot \frac{\partial \text{MSE}}{\partial b_1^2} \\ \text{new } w_{12}^2 &= \text{old } w_{12}^2 - (\text{learning rate}) \cdot \frac{\partial \text{MSE}}{\partial w_{12}^2} \end{aligned}$$





day	temperature (°C)	revenue (in 1000s)
2	22	1.51
6	33	2.22
1	20	1.37
3	25	1.77
2	24	1.64
5	30	2.04
7	35	2.42
4	28	1.90
3	26	1.15
1	21	1.45

gradient descent

① initialize parameters

$b = 0$
 $W_i^n = \#12^n$

② repeat until we reach optimal parameter values:

a) calculate step sizes

(learning rate) $\cdot \frac{\partial \text{MSE}}{\partial b_i^2}$

uses MSE from all data points

b) update all the parameters

new $b_i^2 = \text{old } b_i^2 - \text{step size of } b_i^2$

day	temperature (°C)	revenue (in 1000s)
2	22	1.51
6	33	2.22
1	20	1.37
3	25	1.77
2	24	1.64
5	30	2.04
7	35	2.42
4	28	1.90
3	26	1.15
1	21	1.45

data
point i

Stochastic gradient descent

- ① initialize parameters
- ② repeat until we reach optimal parameter values:

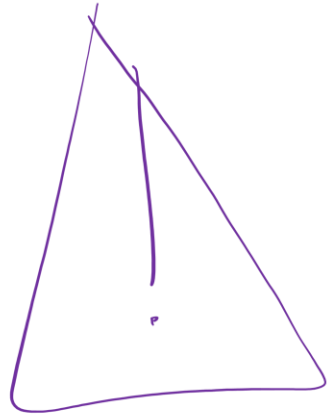
a) calculate step sizes

$$(\text{learning rate}) \cdot \frac{\partial \text{MSE}_i}{\partial b_1^2}$$

use MSE
from
data point i

b) update all the parameters

$$\text{new } b_1^2 = \text{old } b_1^2 - \text{step size of } b_1^2$$



	day	temperature (°C)	revenue (in 1000s)
	2	22	1.51
batch B \	6	33	2.22
	1	20	1.37
	3	25	1.77
	2	24	1.64
	5	30	2.04
	7	35	2.42
	4	28	1.90
	3	26	1.15
	1	21	1.45

mini-batch gradient descent

① initialize parameters

② repeat until we reach optimal parameter values:

a) calculate step sizes

$$(\text{learning rate}) \cdot \frac{1}{B} \sum_{i=1}^B \frac{\partial \text{MSE}}{\partial b_1^2}$$

MSE using the datapoints in the batch

b) update all the parameters

$$\text{new } b_1^2 = \text{old } b_1^2 - \text{step size of } b_1^2$$